

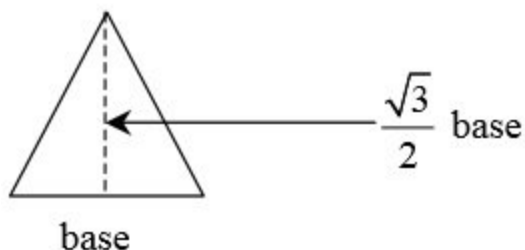
8-8-10

1. Let R be the region enclosed by the graphs of $y = \ln(x^2 + 1)$ and $y = \cos x$.

The base of a solid is the region R . Each cross section of the solid perpendicular to the x -axis is an equilateral triangle. Write an expression involving one or more integrals that gives the volume of the solid. Do not evaluate.

Part C

1 point is earned for the correct evaluation of $V = \int_{-0.916}^{0.916} k [\cos x - \ln(x^2 + 1)]^2 dx$



$$\text{area of cross section} = \frac{1}{2} [\cos x - \ln(x^2 + 1)] \times \left[\frac{\sqrt{3}}{2} (\cos x - \ln(x^2 + 1)) \right]$$

$$\text{volume} = \int_{-B}^B \frac{\sqrt{3}}{4} [\cos x - \ln(x^2 + 1)]^2 dx$$

1 point is earned for the correct constant k

1/2 if sections perpendicular to y -axis $V = \sqrt{3} \int_0^{.609} (e^y - 1) dy + \sqrt{3} \int_{.609}^1 (\cos^{-1} y)^2 dy$

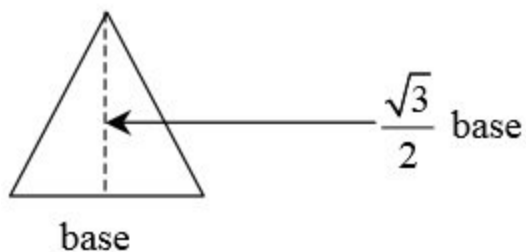


0	1	2
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The student response earns two of the following points:

1 point is earned for the correct evaluation of $V = \int_{-0.916}^{0.916} k [\cos x - \ln(x^2 + 1)]^2 dx$

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$$\text{area of cross section} = \frac{1}{2} [\cos x - \ln(x^2 + 1)] \times \left[\frac{\sqrt{3}}{2} (\cos x - \ln(x^2 + 1)) \right]$$

$$\text{volume} = \int_{-B}^B \frac{\sqrt{3}}{4} [\cos x - \ln(x^2 + 1)]^2 dx$$

1 point is earned for the correct constant k

1/2 if sections perpendicular to y -axis

$$V = \sqrt{3} \int_0^{.609} (e^y - 1) dy + \sqrt{3} \int_{.609}^1 (\cos^{-1} y)^2 dy$$

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2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

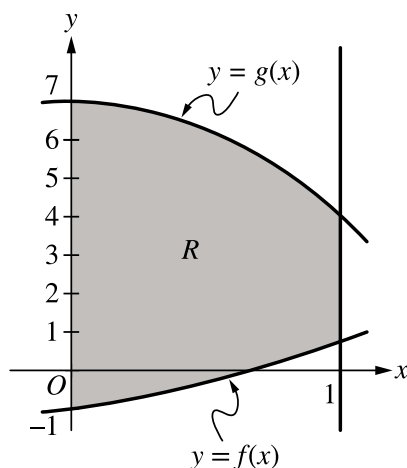


Figure 1

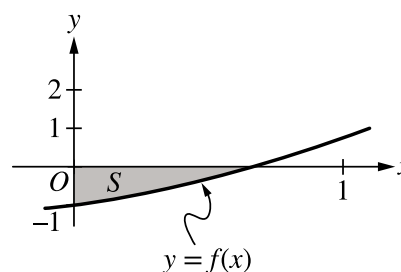


Figure 2

Let f and g be the functions given by $f(x) = e^x - 2$ and $g(x) = -3x^2 + 7$. Let R be the shaded region bounded by the graphs of f and g , the y -axis, and the vertical line $x = 1$, as shown in Figure 1. Let S be the shaded region in Quadrant IV bounded by the graph of f , the x -axis, and the y -axis, as shown in Figure 2.

- Find the area of R .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 8$.
- The region S is the base of a solid. For this solid, each cross section perpendicular to the y -axis is a rectangle with height 5 and base in region S . Write, but do not evaluate, an integral expression that gives the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $g(x) - f(x)$, $f(x) - g(x)$, $|g(x) - f(x)|$, or $|f(x) - g(x)|$ regardless of any subsequent copy errors in presentation.

- A response that presents any subsequent copy or parentheses errors is eligible for the second and third points, but not the fourth.

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A response earns the fourth point only if it has earned the first 3 points and presents an answer equivalent to $9 - e$.

A response that uses an integrand of $f(x) - g(x)$:

- can earn the fourth point if it presents correct communication, such as $\int_0^1 (f(x) - g(x)) \, dx = e - 9$, so the area is $9 - e$.
- does not earn the fourth point if it includes incorrect communication, such as $\int_0^1 (f(x) - g(x)) \, dx = 9 - e$.

Special Case: A response that does not earn the first point, but presents $\int_0^1 (-3x^2 + 7 - e^x - 2) \, dx$ (or equivalent), is eligible for the second and third points, and earns the fourth point only for an answer of $5 - e$.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative of $3x^2$ or e^x
- ☐ Remaining antiderivatives
- ☐ Answer

Solution:

$$\begin{aligned}
 \text{Area} &= \int_0^1 (g(x) - f(x)) \, dx \\
 &= \int_0^1 ((-3x^2 + 7) - (e^x - 2)) \, dx = \int_0^1 (-3x^2 + 9 - e^x) \, dx \\
 &= [-x^3 + 9x - e^x]_0^1 = (-1 + 9 - e) - (-1) = 9 - e
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting both the correct constant and correct limits of integration in an integral expression involving both $f(x)$ and $g(x)$.

The second point is earned by an integrand of the form $R^2 - r^2$, where one of R and r is of the form $(\pm a \pm f(x))$ and the other is of the form $(\pm a \pm g(x))$, for $a \neq 0$.

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The third point is earned only for the correct integrand.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits and constant
- ☐ Form of integrand
- ☐ Integrand

Solution:

$$\text{Volume} = \pi \int_0^1 \left((8 - f(x))^2 - (8 - g(x))^2 \right) dx$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Alternate solution:

$$f(x) = 0 \Rightarrow x = \ln 2$$

$$\text{Volume} = 5 \cdot \text{Area}(S) = 5 \int_0^{\ln 2} (-f(x)) dx$$

These responses earn 1 of the 2 points:

- $5 \int_{-1}^0 \ln(y + 2) dx$
- $5 \int_0^{\ln 2} (-f(x)) dy,$
- $5 \int_0^{\ln 2} -(e^x - 2) dy$
- $5 \int_{-1}^0 \ln(y + 2)$
- $5 \int_0^{\ln 2} -f(x), 5 \int_0^{\ln 2} -(e^x - 2)$
- $5 \int_{-1}^0 -f(x) dy$

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$$\bullet 5 \int_0^{\ln 2} \ln(y+2) \, dx$$



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Limits and constant

☐ Integrand

Solution:

$$y = f(x) = e^x - 2 \Rightarrow x = \ln(y+2)$$

$$\text{Volume} = \int_{-1}^0 5 \ln(y+2) \, dy$$

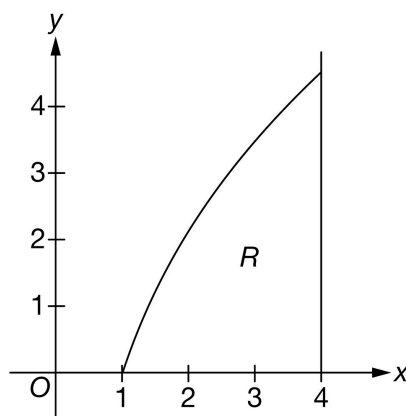
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3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region bounded by the graph of $y = \frac{3(x-1)}{\sqrt{x}}$, the x -axis, and the vertical line $x = 4$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.

Part A

The first point is for the integrand only. At most 3 out of 4 points earned for a copy error in the integrand. Subsequent points may be earned.

The second and third points are for antiderivatives. At most 1 out of 2 antiderivative points earned if only one antiderivative is correct OR there is at most one arithmetic sign error in the antiderivatives.

The fourth point is for the limits of integration and evaluation.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

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- ☐ integrand
- ☐ antiderivatives (no criteria)
- ☐ answer

Solution:

$$\begin{aligned}\text{Area} &= \int_1^4 \frac{3(x-1)}{\sqrt{x}} dx = \int_1^4 \left(3\sqrt{x} - \frac{3}{\sqrt{x}} \right) dx \\ &= \left(3 \cdot \frac{2}{3} x^{\frac{3}{2}} - 3 \cdot 2x^{\frac{1}{2}} \right) \bigg|_{x=1}^{x=4} = \left(2x^{\frac{3}{2}} - 6\sqrt{x} \right) \bigg|_{x=1}^{x=4} \\ &= (16 - 12) - (2 - 6) = 8\end{aligned}$$

Part B

The first point is for the integrand only. At most 4 out of 5 points earned for a copy error in the integrand. Subsequent points may be earned.

The second point is for limits and π .

The third and fourth points are for antiderivatives. At most 1 out of 2 antiderivative points earned if only one antidifferentiation error OR there is at most one arithmetic sign error in the antiderivatives.

The fifth point is for the evaluation, which does not need to be simplified. An error with the factor of 9 applies to this point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4	5
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The student response accurately includes all five of the criteria below.

- ☐ integrand
- ☐ limits and constant
- ☐ antiderivatives (no criteria)
- ☐ answer

Solution:

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$$\begin{aligned}\text{Volume} &= \pi \int_1^4 \left(\frac{3(x-1)}{\sqrt{x}} \right)^2 dx = 9\pi \int_1^4 \left(\frac{x^2 - 2x + 1}{x} \right) dx \\ &= 9\pi \int_1^4 \left(x - 2 + \frac{1}{x} \right) dx = 9\pi \left(\frac{x^2}{2} - 2x + \ln x \right) \Big|_{x=1}^{x=4} \\ &= 9\pi \left((8 - 8 + \ln 4) - \left(\frac{1}{2} - 2 + 0 \right) \right) = 9\pi \left(\ln 4 + \frac{3}{2} \right)\end{aligned}$$

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4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let R be the region in the first quadrant bounded by the graph of $y = \frac{1}{x}$, the horizontal line $y = 1$, and the vertical line $x = e$.

- Find the area of region R .
- Find the volume of the solid generated when region R is revolved about the line $y = 1$.
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a semicircle. Write, but do not evaluate, an expression involving an integral that gives the volume of the solid.

Part A

At most 2 out of 3 points earned if there is one error in the integrand **OR** 1 error with limits of integration. Numerical answers do not need to be simplified (e.g., $\ln e$)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ definite integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\text{Area} = \int_1^e \left(1 - \frac{1}{x}\right) dx = (x - \ln x) \Big|_1^e = (e - \ln e) - (1 - \ln 1) = e - 2$$

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Part B

At most 2 out of 3 points earned if there is one error in the integrand OR 1 error with limits of integration.
Numerical answers do not need to be simplified (e.g., $\ln e$)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ integrand
- ☐ limits and π
- ☐ antiderivative
- ☐ answer

Solution:

$$\begin{aligned}\text{Volume} &= \pi \int_1^e \left(1 - \frac{1}{x}\right)^2 dx = \pi \int_1^e \left(1 - \frac{2}{x} + \frac{1}{x^2}\right) dx \\ &= \pi \left(x - 2 \ln x - \frac{1}{x}\right) \Big|_1^e = \pi \left(e - 2 \ln e - \frac{1}{e}\right) = \pi \left(e - 2 - \frac{1}{e}\right)\end{aligned}$$

Part C

Halving the diameter of the semicircle to find an expression for its radius may appear in the integrand, as presented. It may also appear as part of the constant multiple.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ integrand
- ☐ limits and constant

Solution:

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$$\text{Volume} = \frac{\pi}{2} \int_1^e \left(\frac{1}{2} \left(1 - \frac{1}{x} \right) \right)^2 dx$$

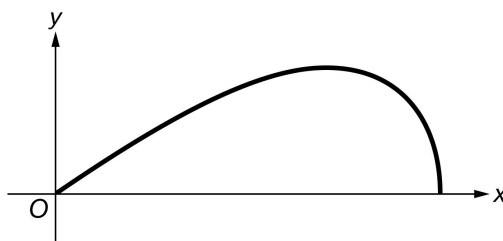
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5. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



A company designs spinning toys using the family of functions $y = cx\sqrt{4 - x^2}$, where c is a positive constant. The figure above shows the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$, for some c . Each spinning toy is in the shape of the solid generated when such a region is revolved about the x -axis. Both x and y are measured in inches.

(a) Find the area of the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$ for $c = 6$.

(b) It is known that, for $y = cx\sqrt{4 - x^2}$, $\frac{dy}{dx} = \frac{c(4 - 2x^2)}{\sqrt{4 - x^2}}$. For a particular spinning toy, the radius of the largest cross-sectional circular slice is 1.2 inches. What is the value of c for this spinning toy?

(c) For another spinning toy, the volume is 2π cubic inches. What is the value of c for this spinning toy?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Units are not required for any points in this question and are not read if presented (correctly or incorrectly) in any part of the response.

The first point is earned for presenting $cx\sqrt{4 - x^2}$ or $6x\sqrt{4 - x^2}$ as the integrand in a definite integral. Limits of integration (numeric or alphanumeric) must be presented (as part of the definite integral) but do not need to be correct in order to earn the first point.

If an indefinite integral is presented with an integrand of the correct form, the first point can be earned if the antiderivative (correct or incorrect) is eventually evaluated using the correct limits of integration.

The second point can be earned without the first point. The second point is earned for the presentation of a correct antiderivative of a function of the form $Ax\sqrt{4 - x^2}$, for any nonzero constant A . If the response has subsequent

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errors in simplification of the antiderivative or sign errors, the response will earn the second point but will not earn the third point.

Responses that use u -substitution and have incorrect limits of integration or do not change the limits of integration from x - to u -values are eligible for the second point.

The response is eligible for the third point only if it has earned the second point.

The third point is earned only for the answer 16 or equivalent. In the case where a response only presents an indefinite integral, the use of the correct limits of integration to evaluate the antiderivative must be shown to earn the third point.

The response cannot correct -16 to $+16$ in order to earn the third point; there is no possible reversal here.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative
- ☐ Answer

Solution:

$$6x\sqrt{4-x^2} = 0 \Rightarrow x = 0, x = 2$$

$$\text{Area} = \int_0^2 6x\sqrt{4-x^2} \, dx$$

$$\text{Let } u = 4 - x^2.$$

$$du = -2x \, dx \Rightarrow -\frac{1}{2} du = x \, dx$$

$$x = 0 \Rightarrow u = 4 - 0^2 = 4$$

$$x = 2 \Rightarrow u = 4 - 2^2 = 0$$

$$\begin{aligned} \int_0^2 6x\sqrt{4-x^2} \, dx &= \int_4^0 6 \left(-\frac{1}{2}\right) \sqrt{u} \, du = -3 \int_4^0 u^{1/2} \, du = 3 \int_0^4 u^{1/2} \, du \\ &= 2u^{3/2} \Big|_{u=0}^{u=4} = 2 \cdot 8 = 16 \end{aligned}$$

The area of the region is 16 square inches.

Part B

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for setting $\frac{dy}{dx} = 0$, $\frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0$, or $c(4-2x^2) = 0$.

An unsupported $x = \sqrt{2}$ does not earn the first point.

The second point can be earned without the first point but is earned only for the answer $c = 0.6$ with supporting work.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Sets $\frac{dy}{dx} = 0$
- ☐ Answer

Solution:

The cross-sectional circular slice with the largest radius occurs where $cx\sqrt{4-x^2}$ has its maximum on the interval $0 < x < 2$.

$$\frac{dy}{dx} = \frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0 \Rightarrow x = \sqrt{2}$$

$$x = \sqrt{2} \Rightarrow y = c\sqrt{2}\sqrt{4 - (\sqrt{2})^2} = 2c$$

$$2c = 1.2 \Rightarrow c = 0.6$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting an integrand of the form $A(x\sqrt{4-x^2})^2$ in a definite integral with any limits of integration (numeric or alphanumeric) and any nonzero constant A . Mishandling the c will result in the response being ineligible for the fourth point.

The second point can be earned without the first point. The second point is earned for the limits of integration, $x = 0$ and $x = 2$, and the constant π (but not for 2π) as part of an integral with a correct or incorrect integrand.

If an indefinite integral is presented with the correct constant π , the second point can be earned if the antiderivative (correct or incorrect) is evaluated using the correct limits of integration.

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A response that presents $2 = \int_0^2 \left(cx\sqrt{4-x^2} \right)^2 dx$ earns the first and second points.

The third point is earned for presenting a correct antiderivative of the presented integrand of the form $A(x\sqrt{4-x^2})^2$ for any nonzero A . If there are subsequent errors in simplification of the antiderivative, linkage errors, or sign errors, the response will not earn the fourth point.

The fourth point cannot be earned without the third point. The fourth point is earned only for the correct answer. The expression does not need to be simplified to earn the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Form of the integrand
- ☐ Limits and constant
- ☐ Antiderivative
- ☐ Answer

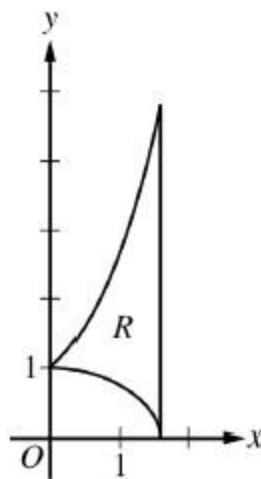
Solution:

$$\begin{aligned}
 \text{Volume} &= \int_0^2 \pi \left(cx\sqrt{4-x^2} \right)^2 dx = \pi c^2 \int_0^2 x^2 (4-x^2) dx \\
 &= \pi c^2 \int_0^2 (4x^2 - x^4) dx = \pi c^2 \left(\frac{4}{3}x^3 - \frac{1}{5}x^5 \right) \Big|_0^2 \\
 &= \pi c^2 \left(\frac{32}{3} - \frac{32}{5} \right) = \frac{64\pi c^2}{15}
 \end{aligned}$$

$$\frac{64\pi c^2}{15} = 2\pi \Rightarrow c^2 = \frac{15}{32} \Rightarrow c = \sqrt{\frac{15}{32}}$$

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6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Let R be the region in the first quadrant enclosed by the graph of $f(x) = \sqrt{\cos x}$, the graph of $g(x) = e^x$, and the vertical line $x = \frac{\pi}{2}$, as shown in the figure above.

- Write, but do not evaluate, an integral expression that gives the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R is the base of a solid whose cross sections perpendicular to the x -axis are semicircles with diameters on the xy -plane. Write, but do not evaluate, an integral expression that gives the volume of this solid.

Part A**1 point is earned for:** integrand**1 point is earned for:** limits

$$\text{Area} = \int_0^{\pi/2} (g(x) - f(x)) dx$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: integrand**1 point is earned for:** limits

$$\text{Area} = \int_0^{\pi/2} (g(x) - f(x)) dx$$

Part B

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1 point is earned for: integrand

2 points are earned for: antiderivatives

1 point is earned for: answer

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\pi/2} \left((g(x))^2 - (f(x))^2 \right) dx \\
 &= \pi \int_0^{\pi/2} \left((e^x)^2 - (\sqrt{\cos x})^2 \right) dx \\
 &= \pi \int_0^{\pi/2} (e^{2x} - \cos x) dx = \pi \left[\frac{1}{2} e^{2x} - \sin x \right]_{x=0}^{x=\pi/2} \\
 &= \pi \left(\frac{1}{2} e^{\pi} - \sin \frac{\pi}{2} - \left(\frac{1}{2} - 0 \right) \right) = \pi \left(\frac{1}{2} e^{\pi} - \frac{3}{2} \right)
 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: integrand

2 points are earned for: antiderivatives

1 point is earned for: answer

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\pi/2} \left((g(x))^2 - (f(x))^2 \right) dx \\
 &= \pi \int_0^{\pi/2} \left((e^x)^2 - (\sqrt{\cos x})^2 \right) dx \\
 &= \pi \int_0^{\pi/2} (e^{2x} - \cos x) dx = \pi \left[\frac{1}{2} e^{2x} - \sin x \right]_{x=0}^{x=\pi/2} \\
 &= \pi \left(\frac{1}{2} e^{\pi} - \sin \frac{\pi}{2} - \left(\frac{1}{2} - 0 \right) \right) = \pi \left(\frac{1}{2} e^{\pi} - \frac{3}{2} \right)
 \end{aligned}$$

Part C

1 point is earned for: integrand

1 point is earned for: limits

1 point is earned for: constant

$$\text{Volume} = \frac{1}{2} \int_0^{\pi/2} \pi \left(\frac{g(x) - f(x)}{2} \right)^2 dx$$

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0	1	2	3
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The student response earns all of the following points:

1 point is earned for: integrand

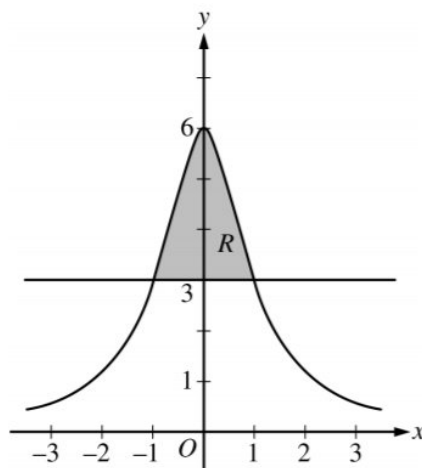
1 point is earned for: limits

1 point is earned for: constant

$$\text{Volume} = \frac{1}{2} \int_0^{\pi/2} \pi \left(\frac{g(x) - f(x)}{2} \right)^2 dx$$

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7. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Let f be the function defined by $f(x) = \frac{6}{1+x^2}$. Let R be the shaded region bounded by the graph of f and the horizontal line $y = 3$, as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

(c) Let $h(x)$ be the vertical distance between the point $(x, f(x))$ and the horizontal line $y = 3$. Find the rate of change of $h(x)$ with respect to x at $x = 2$.

Part A

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) \, dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) \, dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.

8-8-10



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) \, dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) \, dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.

Part B

2 points are earned for: integrand

1 point is earned for: limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) \, dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for: integrand

1 point is earned for: limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) \, dx$$

8-8-10

Part C

1 point is earned for: $h(x)$ 1 point is earned for: $h'(x)$ 1 point is earned for: $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $h(x)$ 1 point is earned for: $h'(x)$ 1 point is earned for: $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$

8-8-10

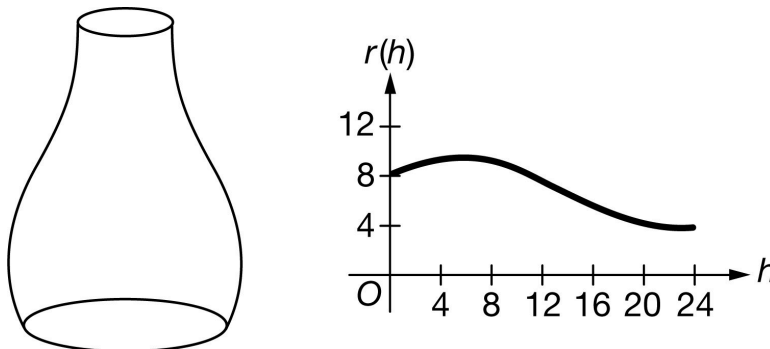
8. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



A 24-centimeter tall vase has cross sections parallel to the base that are circles. Each circular cross section h centimeters above the base has a radius of $r(h) = 8 - 0.1h + 2 \sin(0.3h^{0.9})$ centimeters for $0 \leq h \leq 24$. A sketch of the vase and the graph of r are shown above.

(a) Find the area of the region between the graph of r and the h -axis from $h = 0$ to $h = 24$.

(b) Find the volume of the vase.

(c) Evaluate $\int_4^{20} r'(h) \, dh$. Using correct units, interpret the meaning of $\int_4^{20} r'(h) \, dh$ in the context of the problem.

(d) Water is poured into the vase. At the instant when the depth of the water in the vase is 10 centimeters, the depth of the water is increasing at a rate of 0.6 centimeter per second. At that instant, what is the rate of change of the circumference of the surface of the water, in centimeters per second?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported answer of 165.781 (or 165.78) will not earn any points.

8-8-10

The first point is earned by using $r(h)$ as the integrand in a definite integral. A definite integral with incorrect limits or multiplied by any constant other than 1 will not earn the second point.

Do not read units in this problem.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\text{area} = 165.509$ (or 165.508).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\int_0^{24} r(h) \, dh = 165.780506$$

The area of the region is 165.781 (or 165.78).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for writing a definite integral (with any limits) with integrand $c(r)^2$, for any nonzero constant c .

Once the first point is earned, it cannot be lost. Any subsequent copy errors in the presence of the correct answer will be considered a calculator restart and will earn the second point.

To earn the second point, a response must earn the first point, show limits from 0 to 24, multiply by π , and present the correct evaluation.

Special case: A response that fails to earn the first point because of a copy error will earn the second point for the correct answer.

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the second point. (For example, 1252.387π is not correct to three digits after the decimal and therefore does not earn the second point.)

Degree mode: $\text{volume} = 3616.746$ (or 3616.745) (See degree mode statement in part (a).)

8-8-10



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\int_0^{24} \pi(r(h))^2 dh = \pi \cdot 1252.387790 = 3934.492281$$

The volume of the vase is 3934.492 centimeters.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the value -5.259 ; no supporting work is required.

The interpretation must clearly indicate that the radius at $h = 20$ is smaller than the radius at $h = 4$. Therefore, “the difference in the radius when $h = 20$ and $h = 4$...” is not sufficient.

The response, “The difference (or change) in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns both points. However, “The *total* change in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns only the first point.

Degree mode: answer = -1.481 (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation with units

Solution:

$$\int_4^{20} r'(h) dh = r(20) - r(4) = -5.259388$$

8-8-10

The radius of the vase at $h = 20$ centimeters is 5.259 centimeters less than the radius of the vase at $h = 4$ centimeters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by using $\frac{dr}{dh} \cdot \frac{dh}{dt} \frac{dr}{dt}$ in an expression for $\frac{dC}{dt}$.

A response may work from $C = 2\pi(8 - 0.1h + 2\sin(0.3h^{0.9}))$. In this case, the first point is earned for $\frac{dC}{dt} = \frac{dC}{dh} \cdot \frac{dh}{dt}$.

The second point is earned for using either the value 0.6 as $\frac{dh}{dt}$ or the value -0.411320 as $\frac{dr}{dh}$ in an expression for $\frac{dC}{dt}$. This point is not earned if either 0.6 or $r'(10)$ is used as $\frac{dr}{dt}$.

The second point can be earned without the first point. For example, a computation or communication error in attempting to use the chain rule for $\frac{dC}{dt}$ does not earn the first point but is eligible for the second point.

A response that uses an incorrect equation for the circumference in terms of r only, such as $C = \pi r^2$, is eligible for the first two points but not the third point.

The third point is earned only for an answer of -1.551 (or -1.55).

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the third point.

Degree mode: answer = -0.349 (or -0.348) (See degree mode statement in part (a).)



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\frac{dr}{dt} = \frac{dr}{dh} \cdot \frac{dh}{dt}$
- Uses $\frac{dh}{dt} \Big|_{h=10} = 0.6$
- ☐ – OR –
- Uses $\frac{dr}{dh} \Big|_{h=10} = r'(10)$
- ☐ Answer

Solution:

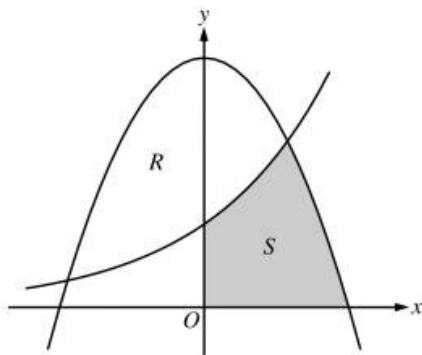
8-8-10

$$C = 2\pi r \Rightarrow \frac{dC}{dt} = \frac{d}{dt}(2\pi r) = 2\pi \cdot \left(\frac{dr}{dh} \cdot \frac{dh}{dt} \right)$$

$$\begin{aligned} \frac{dC}{dt} \Big|_{h=10} &= 2\pi \cdot \left(\frac{dr}{dh} \Big|_{h=10} \cdot \frac{dh}{dt} \Big|_{h=10} \right) \\ &= 2\pi \cdot (r'(10) \cdot 0.6) = \pi \cdot (-0.493584) = -1.550641 \end{aligned}$$

The rate of change of the circumference of the surface of the water is -1.551 (or -1.55) centimeters per second.

8-8-10

9. 

Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

- Find the area of S .
- Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

Let $a = -1.63658$

$$\begin{aligned} \text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrands

8-8-10

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

Let $a = -1.63658$

$$\begin{aligned} \text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$

Part B

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107 \end{aligned}$$

Part C

1 point is earned for integrand

8-8-10

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$



0	1	2
---	---	---

The student response earns all of the following points:

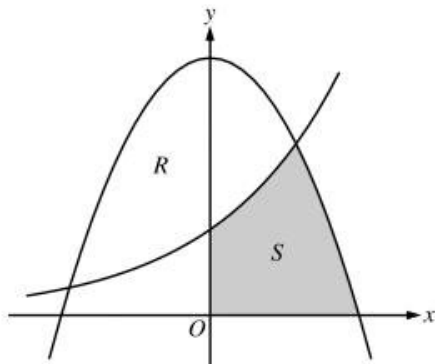
1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

8-8-10

10.



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part C

1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$



0	1	2
---	---	---

The student response earns all of the following points:

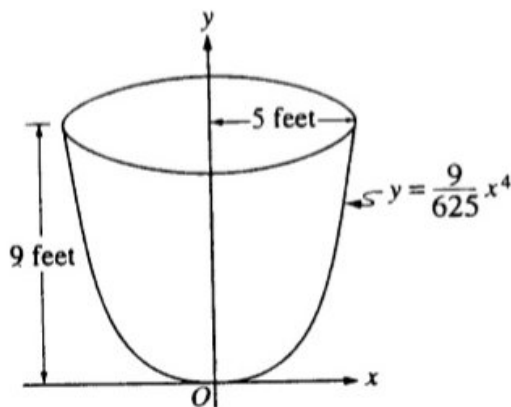
1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

8-8-10

11.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate all units of measure.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

8-8-10



0	1	2	3	4	5
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The student response earns five of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

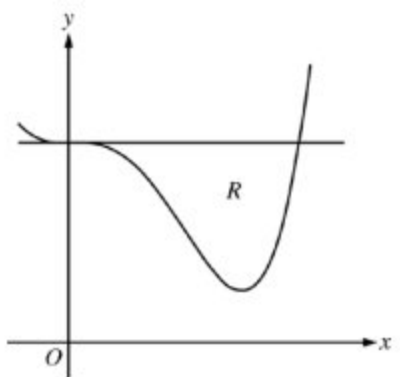
$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

8-8-10

12.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

 Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in R . Find the volume of the solid.

Part B

2 points are earned for integrand

1 point is earned for answer

$$\text{Volume} = \int_0^{2.3} \frac{1}{2} (4 - f(x)) 2 \, dx$$

$$= 3.574 \text{ (or } 3.573)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

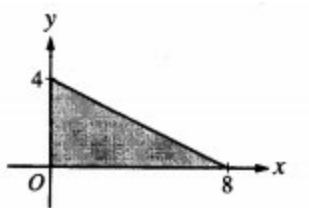
1 point is earned for answer

$$\text{Volume} = \int_0^{2.3} \frac{1}{2} (4 - f(x)) 2 \, dx$$

$$= 3.574 \text{ (or } 3.573)$$

8-8-10

13.



The base of a solid is a region in the first quadrant bounded by the x -axis, the y -axis, and the line $x+2y=8$, as shown in the figure above. If cross sections of the solid perpendicular to the x -axis are semicircles, what is the volume of the solid?

- (A) 12.566
(B) 14.661
(C) 16.755
(D) 67.021
(E) 134.041



8-8-10

14. Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.



The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles. Find the volume of this solid.

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

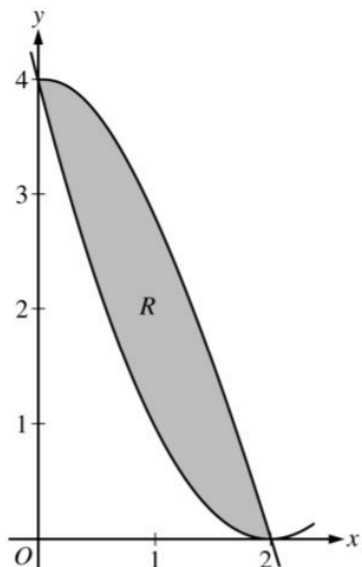
2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268 \end{aligned}$$

8-8-10

15.



Let R be the region in the first quadrant bounded by the graphs of $y = 4 \cos(\frac{\pi x}{4})$ and $y = (x - 2)^2$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in region R .



What is the volume of the solid?

(A) 1.775

(B) 3.549

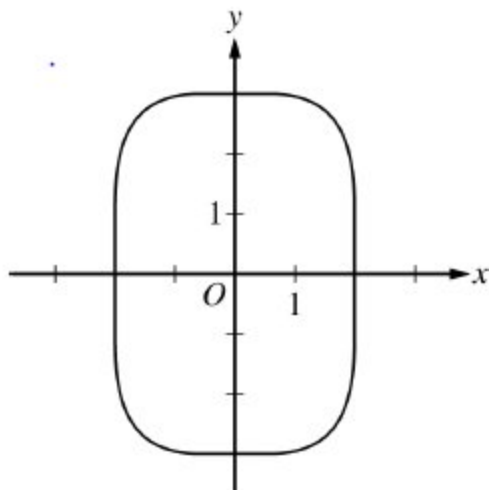
(C) 4.800

(D) 5.575



8-8-10

16.



The base of a solid is the region enclosed by the curve $\frac{x^4}{16} + \frac{y^4}{81} = 1$ shown in the figure above. For the solid, each cross section perpendicular to the x -axis is a semicircle. What is the volume of the solid?

- (A) 12.356
- (B) 15.732
- (C) 22.249
- (D) 24.712
- (E) 49.425



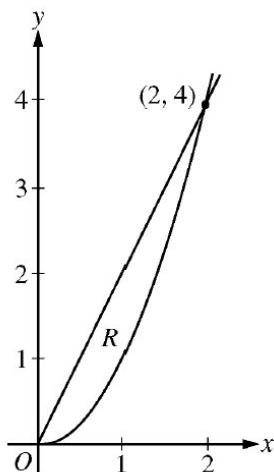
17. The region bounded by the graph of $y = 2x - x^2$ and the x -axis is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an equilateral triangle. What is the volume of the solid?

- (A) 1.333
- (B) 1.067
- (C) 0.577
- (D) 0.462
- (E) 0.267



8-8-10

18.



Let R be the region in the first quadrant enclosed by the graphs of $y=2x$ and $y=x^2$, as shown in the figure above.

The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis has area $A(x)=\sin(\pi/2x)$. Find the volume of the solid.

Part B

1 point is earned for the integrand

1 point is earned for the anti derivative

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi} \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the integrand

1 point is earned for the anti derivative

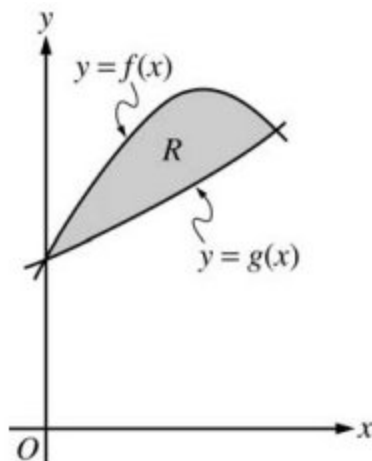
1 point is earned for the answer

8-8-10

$$\begin{aligned} \text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi} \end{aligned}$$

8-8-10

19.



Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure above.

 The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles with diameters extending from $y=f(x)$ to $y=g(x)$. Find the volume of this solid.

Part C

2 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^s \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^s \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 \end{aligned}$$

=0.077 or 0.078



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for the integrand

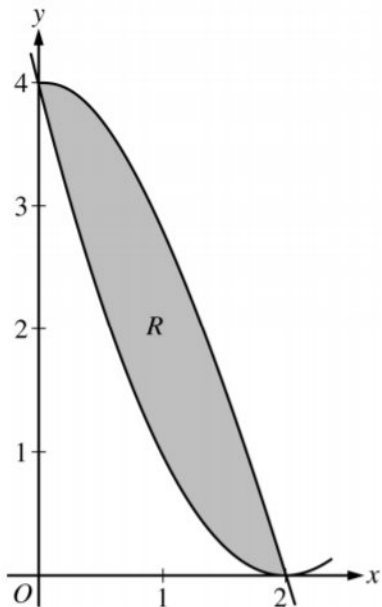
1 point is earned for the answer

8-8-10

$$\begin{aligned} \text{Volume} &= \int_0^s \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^s \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \end{aligned}$$

$$= 0.077 \text{ or } 0.078$$

20.



Let R be the region in the first quadrant bounded by the graphs of $y = 4 \cos\left(\frac{\pi x}{4}\right)$ and $y = (x-2)^2$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in region R .



What is the volume of the solid?

(A) 1.775

(B) 3.549

(C) 4.800

(D) 5.575



8-8-10

21. Let R be the region enclosed by the graph of $y = \sqrt{x - 1}$, the vertical line $x=10$, and the x -axis.



Find the volume of the solid generated when R is revolved about the vertical line $x=10$.

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$=407.150$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$=407.150$$

22. A vase has the shape obtained by revolving the curve $y = 2 + \sin x$ from $x = 0$ to $x = 5$ about the x -axis, where x and y are measured in inches. What is the volume, in cubic inches, of the vase?

(A) 10.716

(B) 25.501

(C) 33.666

(D) 71.113

(E) 80.115



8-8-10

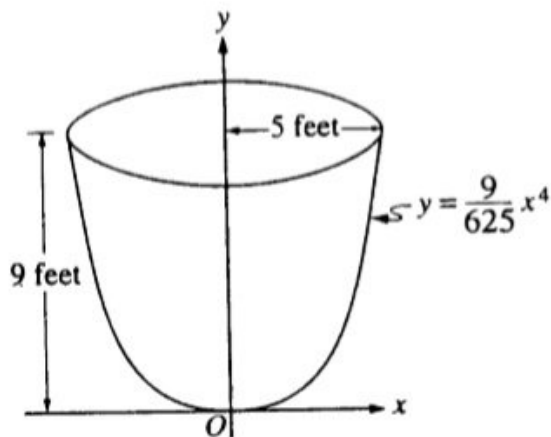
23.  What is the volume of the solid generated when the region bounded by the graph of $x = \sqrt{y - 2}$ and the lines $x = 0$ and $y = 5$ is revolved about the y -axis?

- (A) 3.464
(B) 4.500
(C) 7.854
(D) 10.883
(E) 14.137



8-8-10

24.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x=0$ to $x=5$ about the y -axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.

Find the volume of the tank. Indicate units of measure.

Part A

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \pi \int_0^9 () dy, 2\pi \int_0^5 () dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi \text{ ft}^3$$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi \int_0^5 x \left(9 - \frac{9}{625}x^4 \right) dx = 150\pi \text{ ft}^3$$

(or 471.238 ft^3 or 471.239 ft^3)

[465, 475] [148π, 152π]

8-8-10



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \int_0^9 () dy, 2\pi \int_0^5 () dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi ft^3$$

(or $471.238 ft^3$ or $471.239 ft^3$)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi \int_0^5 x \left(9 - \frac{9}{625} x^4 \right) dx = 150\pi ft^3$$

(or $471.238 ft^3$ or $471.239 ft^3$)

[465, 475] [148π, 152π]

25. The region in the first quadrant bounded by the graph of $y = \sec x$, $x = \frac{\pi}{4}$, and the axes is rotated about the x -axis. What is the volume of the solid generated?

(A) $\frac{\pi^2}{4}$

(B) $\pi - 1$

(C) π

(D) 2π

(E) $\frac{8\pi}{3}$



8-8-10

26. The region in the first quadrant bounded by the graph of $y = \sec x$, $x = \frac{\pi}{4}$, and the axes is rotated about the x -axis. What is the volume of the solid generated?

- (A) $\frac{\pi^2}{4}$
(B) $\pi - 1$
(C) π
(D) 2π
(E) $\frac{8\pi}{3}$



27.  What is the volume of the solid generated by rotating about the x -axis the region enclosed by the curve $y = \sec x$ and the lines $x=0$, $y=0$, and $x = \frac{\pi}{3}$?

- (A) $\frac{\pi}{\sqrt{3}}$
(B) π
(C) $\pi\sqrt{3}$
(D) $\frac{8\pi}{3}$
(E) $\pi \ln\left(\frac{1}{2} + \sqrt{3}\right)$



28. The region R in the first quadrant is enclosed by the lines $x=0$ and $y=5$ and the graph of $y = x^2 + 1$. The volume of the solid generated when R is revolved about the y -axis is

- (A) 6π
(B) 8π
(C) $34\pi/3$
(D) 16π
(E) $544\pi/15$



8-8-10

Let R be the region bounded by the x -axis, the graph of $y = \sqrt{x}$, and the line $x = 4$.

29. Find the volume of the solid generated when R is revolved about the x -axis.

Part C

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132

8-8-10

30. The vertical line $x = k$ divides the region R into two regions such that when these two regions are revolved about the x -axis, they generate solids with equal volumes. Find the value of k

Part D

1 point is earned for equation in k

1 point is earned for the answer

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi$$

$$\pi \frac{k^2}{2} = 4\pi$$

OR

$$\pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

$$\pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \text{ or } 2.828$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for equation in k

1 point is earned for the answer

8-8-10

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi$$

$$\pi \frac{k^2}{2} = 4\pi$$

OR

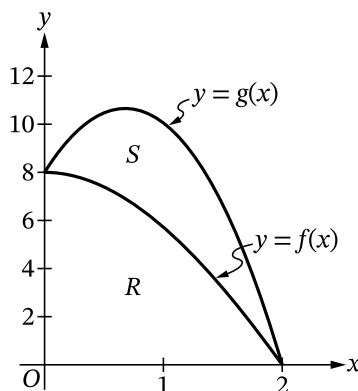
$$\pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

$$\pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \text{ or } 2.828$$

8-8-10

31. The function f is defined by $f(x) = 8 \cos\left(\frac{\pi}{4}x\right)$, and g is a continuous function such that $\int_0^2 g(x) dx = 16$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ for $0 \leq x \leq 2$ and the x - and y -axes. Let S be the region in the first quadrant bounded by the graphs of $y = f(x)$ and $y = g(x)$ for $0 \leq x \leq 2$. Regions R and S are shown in the figure.

**Part A**

Find the area of region R . Show the work that leads to your answer.

Part B

Find the area of region S .

Part C

Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a triangle with its base in region R and with height equal to the length of its base. Write, but do not evaluate, an integral expression for the volume of the solid.

Part D

For $y = f(x)$ on the interval $0 \leq x \leq 2$, it can be shown that $x = \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)$ for $0 \leq y \leq 8$. Write, but do not evaluate, an integral expression that gives the volume of the solid when R is rotated about the vertical line $x = 3$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a definite integral with an integrand of either $f(x)$ or $8 \cos\left(\frac{\pi}{4}x\right)$, with or without the differential dx . A definite integral with incorrect limits of integration is eligible to earn **P1**.

8-8-10

Model Solution:

$$\text{Area of } R = \int_0^2 f(x) dx$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A definite integral with incorrect limits of integration is eligible to earn **P2**.
- **P2** is earned for an antiderivative of the form $k \sin\left(\frac{\pi}{4}x\right)$ or equivalent, where k is any positive constant.

Model Solution:

$$\int_0^2 8 \cos\left(\frac{\pi}{4}x\right) dx = \frac{32}{\pi} \sin\left(\frac{\pi}{4}x\right) \Big|_0^2$$



0	1
---	---

The response accurately includes the criteria below.

- Form of antiderivative

Part A Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response earns **P3** if it has earned **P1** and **P2**, presents a definite integral with correct limits, and reports an answer equivalent to $\frac{32}{\pi}$.

Model Solution:

8-8-10

The area of region R is $\frac{32}{\pi}(\sin(\frac{\pi}{2}) - \sin 0) = \frac{32}{\pi}$.



0	1
---	---

The response accurately includes the criteria below.

- Area of R

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for an answer of $16 - [\text{answer from part A}]$, with no supporting work.

Model Solution:

$$\text{Area of } S = 16 - \frac{32}{\pi}$$



0	1
---	---

The response accurately includes the criteria below.

- Area of S

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P5** is earned only for a response with an integrand of the form $k(f(x))^2$ in a definite or indefinite integral, where k is any nonzero constant.

Model Solution:

$$\begin{aligned} \text{Volume} &= \int_0^2 \frac{1}{2}(f(x))^2 dx \\ &= \int_0^2 \frac{1}{2} \left(8 \cos\left(\frac{\pi}{4}x\right) \right)^2 dx = 32 \int_0^2 \cos^2\left(\frac{\pi}{4}x\right) dx \end{aligned}$$

8-8-10



0	1
---	---

The response accurately includes the criteria below.

- Form of integrand

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P6** is earned for the correct definite integral, with or without the differential dx .
- Any of $\int_0^2 \frac{1}{2}(f(x))^2 dx$, $\int_0^2 \frac{1}{2}\left(8 \cos\left(\frac{\pi}{4}x\right)\right)^2 dx$, $32 \int_0^2 \cos^2\left(\frac{\pi}{4}x\right) dx$, or a mathematically equivalent expression is sufficient to earn **P6**.

Model Solution:

$$\begin{aligned}\text{Volume} &= \int_0^2 \frac{1}{2}(f(x))^2 dx \\ &= \int_0^2 \frac{1}{2}\left(8 \cos\left(\frac{\pi}{4}x\right)\right)^2 dx = 32 \int_0^2 \cos^2\left(\frac{\pi}{4}x\right) dx\end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for a response that presents an integrand of $3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2$, $3^2 - \left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2$, $\left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2$, $\left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2 - 3^2$, or any mathematically equivalent expression, in one or more definite or indefinite integrals.

Model Solution:

8-8-10

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2 \right) dy$$



0	1
---	---

The response accurately includes the criteria below.

- Form of integrand

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned only for an integrand mathematically equivalent to $3^2 - \left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2$ or $3^2 - \left(\frac{4}{\pi} \arccos \left(\frac{y}{8} \right) - 3 \right)^2$ in one or more definite or indefinite integral.
 - For example: $\int_0^8 \left| \left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2 - 3^2 \right| dy$ or $\int_0^8 \left(\left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2 - 3^2 \right) dy$ earns **P8**.
 - Note: An integrand of $\left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2 - 3^2$ or $\left(\frac{4}{\pi} \arccos \left(\frac{y}{8} \right) - 3 \right)^2 - 3^2$ does not earn **P8** unless the reversal is resolved.

Model Solution:

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi} \arccos \left(\frac{y}{8} \right) \right)^2 \right) dy$$



0	1
---	---

The response accurately includes the criteria below.

- Integrand

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

8-8-10

- To be eligible for **P9**, a response must have earned **P7** or **P8** or must have presented an integrand involving $\frac{4}{\pi}\arccos\left(\frac{y}{8}\right)$ of the form $R^2 - r^2$ in a definite integral.
- P9** is earned only by an eligible definite integral with an integrand in terms of y , including the constant π , limits $y = 0$ to $y = 8$, and differential dy . A response that presents any other constant, limits (including $y = 8$ to $y = 0$, except in the note below), or differential does not earn **P9**.

◦ Note: $\pi \int_8^0 \left(\left(3 - \frac{4}{\pi}\arccos\left(\frac{y}{8}\right) \right)^2 - 3^2 \right) dy$ earns **P9**.

- The integral may be split into two integrals; for example, $\pi \int_0^8 3^2 dy - \pi \int_0^8 \left(3 - \frac{4}{\pi}\arccos\left(\frac{y}{8}\right) \right)^2 dy$ or

$$\pi \cdot 3^2 \cdot 8 - \pi \int_0^8 \left(3 - \frac{4}{\pi}\arccos\left(\frac{y}{8}\right) \right)^2 dy.$$

Model Solution:

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi}\arccos\left(\frac{y}{8}\right) \right)^2 \right) dy$$



0	1
---	---

The response accurately includes the criteria below.

- Limits, constant, and differential

8-8-10

32. The base of a solid is the region in the first quadrant between the graph of $y = 2x$ and the x -axis for $0 \leq x \leq 1$. For the solid, each cross section perpendicular to the x -axis is a quarter circle with the corresponding circle's center on the x -axis and one radius in the xy -plane. What is the volume of the solid?

(A) $\frac{\pi}{3}$



(B) $\frac{4}{3}$

(C) $\frac{\pi}{4}$

(D) 1

Answer A

Correct. The area of a quarter circle with radius y is $\frac{\pi}{4}y^2$. At each x , the cross section perpendicular to the x -axis has radius $y = 2x$ and area $\frac{\pi}{4}(2x)^2 = \pi x^2$. Therefore, the volume of the solid is given by the definite integral

$\int_0^1 \pi x^2 \, dx$, evaluated as follows.

$$\int_0^1 \pi x^2 \, dx = \pi \frac{x^3}{3} \bigg|_0^1 = \frac{\pi}{3} - 0 = \frac{\pi}{3}$$

33. Let R be the region in the first quadrant bounded by the graph of $y = x^3$, the line $x = 2$, and the x -axis. R is the base of a solid whose cross sections perpendicular to the x -axis are equilateral triangles. What is the volume of the solid?

(A) $\frac{2}{3}\sqrt{3}$

(B) $\frac{32}{7}\sqrt{3}$



(C) $\frac{64}{7}$

(D) $\frac{128}{7}\pi$

Answer B

Correct. The area of an equilateral triangle with side length s is $\frac{\sqrt{3}}{4}s^2$. A typical cross section of the solid is an equilateral triangle with side length $s = x^3$ and which therefore has area $\frac{\sqrt{3}}{4}(x^3)^2 = \frac{\sqrt{3}}{4}x^6$. The volume of the solid is found using the definite integral of the cross-sectional area, as follows.

$$\int_0^2 \frac{\sqrt{3}}{4}x^6 \, dx = \frac{\sqrt{3}}{4} \cdot \frac{1}{7}x^7 \bigg|_0^2 = \frac{\sqrt{3}}{4} \cdot \frac{128}{7} = \frac{32}{7}\sqrt{3}$$

8-8-10

34. The base of a solid is the region in the first quadrant between the graph of $y = x^2$ and the x -axis for $0 \leq x \leq 1$. For the solid, each cross section perpendicular to the x -axis is a quarter circle with the corresponding circle's center on the x -axis and one radius in the xy -plane. What is the volume of the solid?

(A) $\frac{\pi}{20}$



(B) $\frac{1}{5}$

(C) $\frac{\pi}{12}$

(D) $\frac{1}{3}$

Answer A

Correct. The area of a quarter circle with radius y is $\frac{\pi}{4}y^2$. At each x , the cross section perpendicular to the x -axis has radius $y = x^2$ and area $\frac{\pi}{4}(x^2)^2 = \frac{\pi}{4}x^4$. Therefore, the volume of the solid is given by the definite integral

$\int_0^1 \frac{\pi}{4}x^4 \, dx$, evaluated as follows.

$$\int_0^1 \frac{\pi}{4}x^4 \, dx = \left. \frac{\pi}{20}x^5 \right|_0^1 = \frac{\pi}{20}$$

35. The base of a solid is the region in the first quadrant between the graph of $y = x^2$ and the x -axis for $0 \leq x \leq 1$. For the solid, each cross section perpendicular to the x -axis is a semicircle. What is the volume of the solid?

(A) $\frac{\pi}{40}$



(B) $\frac{\pi}{24}$

(C) $\frac{\pi}{12}$

(D) $\frac{\pi}{10}$

Answer A

Correct. The area of a semicircle with diameter y is $\frac{\pi}{2}\left(\frac{y}{2}\right)^2 = \frac{\pi}{8}y^2$. At each x , the cross section perpendicular to the x -axis has diameter $y = x^2$ and area $\frac{\pi}{8}y^2 = \frac{\pi}{8}(x^2)^2 = \frac{\pi}{8}x^4$. Therefore, the volume of the solid is given by the definite integral $\int_0^1 \frac{\pi}{8}x^4 \, dx$, evaluated as follows.

$$\int_0^1 \frac{\pi}{8}x^4 \, dx = \left. \frac{\pi}{40}x^5 \right|_0^1 = \frac{\pi}{40}$$

8-8-10

36. Let R be the region in the first quadrant bounded by the graph of $y = x^2$, the line $x = 4$, and the x -axis. R is the base of a solid whose cross sections perpendicular to the x -axis are equilateral triangles. What is the volume of the solid?
- (A) $\frac{16}{3}\sqrt{3}$
- (B) $\frac{256}{5}\sqrt{3}$ ✓
- (C) $\frac{512}{5}$
- (D) $\frac{1024}{5}\pi$

Answer B

Correct. The area of an equilateral triangle with side length s is $\frac{\sqrt{3}}{4}s^2$. A typical cross section of the solid is an equilateral triangle with side length $s = x^2$ and which therefore has area $\frac{\sqrt{3}}{4}(x^2)^2 = \frac{\sqrt{3}}{4}x^4$. The volume of the solid is found using the definite integral of the cross-sectional area, as follows.

$$\int_0^4 \frac{\sqrt{3}}{4}x^4 dx = \frac{\sqrt{3}}{4} \cdot \frac{x^5}{5} \Big|_0^4 = \frac{1024\sqrt{3}}{4(5)} = \frac{256}{5}\sqrt{3}$$

37.  The base of a solid is the region in the first quadrant bounded by the graph of $y = \cos x$ and the x - and y -axes for $0 \leq x \leq \frac{\pi}{2}$. For the solid, each cross section perpendicular to the x -axis is an equilateral triangle. What is the volume of the solid?
- (A) 0.340 ✓
- (B) 0.433
- (C) 0.785
- (D) 1.000

Answer A

Correct. The area of an equilateral triangle with side y is $\frac{\sqrt{3}}{4}y^2$. At each x , the cross section perpendicular to the x -axis has side $y = \cos x$ and area $\frac{\sqrt{3}}{4}\cos^2 x$. Therefore, the volume of the solid is given by the definite integral $\int_0^{\frac{\pi}{2}} \frac{\sqrt{3}}{4}\cos^2 x dx$, which can be evaluated using the graphing calculator.

8-8-10

38. Let R be the region in the first quadrant, bounded by the graph of $y = \sqrt{x}$, the y -axis, and the line $y = 1$. The region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is an isosceles right triangle with the right angle on the y -axis and one leg in the xy -plane. What is the volume of the solid?

(A) $\frac{1}{10}$

(B) $\frac{1}{6}$

(C) $\frac{1}{5}$

(D) $\frac{1}{4}$

**Answer A**

Correct. The triangle is bounded on the right by the curve $x = y^2$ for $0 \leq y \leq 1$. The area of an isosceles right triangle with leg h is $\frac{1}{2}h^2$. At each y , the cross section perpendicular to the y -axis has leg $x = y^2$ and area

$\frac{1}{2}(y^2)^2 = \frac{y^4}{2}$. Therefore, the volume of the solid is given by the definite integral $\int_0^1 \frac{y^4}{2} dy$, evaluated as follows.

$$\int_0^1 \frac{y^4}{2} dy = \frac{1}{2} \left(\frac{y^5}{5} \Big|_0^1 \right) = \frac{1}{2} \left(\frac{1}{5} - 0 \right) = \frac{1}{10}$$

39. Let R be the region in the first quadrant bounded by the graph of $y = 4e^x + 2$, the line $x = 1$, and the x - and y -axes. R is the base of a solid whose cross sections perpendicular to the x -axis are semicircles. What is the volume of the solid?

(A) $\pi \left(e^2 + 2e - \frac{5}{2} \right)$

(B) $\pi (2e^2 + 4e - 5)$

(C) $\pi (4e^2 + 8e - 10)$

(D) $\pi (8e^2 + 16e - 20)$

**Answer A**

Correct. The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. A typical cross section of the solid is a semicircle with diameter from the x -axis to the graph of $y = 4e^x + 2$. The length of the diameter is $4e^x + 2$, and therefore the radius of the semicircle is $r = \frac{4e^x + 2}{2} = 2e^x + 1$. The volume of the solid is found using the definite integral of the cross-sectional area as follows.

$$\begin{aligned} \frac{\pi}{2} \int_0^1 (2e^x + 1)^2 dx &= \frac{\pi}{2} \int_0^1 (4e^{2x} + 4e^x + 1) dx = \frac{\pi}{2} (2e^{2x} + 4e^x + x) \Big|_0^1 \\ &= \frac{\pi}{2} (2e^2 + 4e + 1 - (2 + 4)) = \pi \left(e^2 + 2e - \frac{5}{2} \right) \end{aligned}$$

8-8-10

40. Let R be the triangular region in the first quadrant, with vertices at points $(0, 0)$, $(0, 2)$, and $(1, 2)$. The region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is an isosceles right triangle with the right angle on the y -axis and one leg in the xy -plane. What is the volume of the solid?

- (A) $\frac{1}{3}$
(B) $\frac{1}{2}$
(C) $\frac{2}{3}$
(D) $\frac{16}{3}$

**Answer A**

Correct. The triangle is bounded on the right by the line $x = \frac{y}{2}$ for $0 \leq y \leq 2$. The area of an isosceles right triangle with leg x is $\frac{x^2}{2}$. At each y , the cross section perpendicular to the y -axis has leg $x = \frac{y}{2}$ and area $\frac{(\frac{y}{2})^2}{2} = \frac{y^2}{8}$. Therefore, the volume of the solid is given by the definite integral $\int_0^2 \frac{y^2}{8} dy$, evaluated as follows.

$$\int_0^2 \frac{y^2}{8} dy = \frac{y^3}{24} \Big|_0^2 = \frac{8}{24} - 0 = \frac{1}{3}$$

41. The base of a solid is the region in the first quadrant between the graph of $y = x^3$ and the x -axis for $0 \leq x \leq 1$. For the solid, each cross section perpendicular to the x -axis is a semicircle. What is the volume of the solid?

- (A) $\frac{\pi}{56}$
(B) $\frac{\pi}{32}$
(C) $\frac{\pi}{16}$
(D) $\frac{\pi}{14}$

**Answer A**

Correct. The area of a semicircle with diameter y is $\frac{\pi}{2} \left(\frac{y}{2}\right)^2 = \frac{\pi}{8} y^2$. At each x , the cross section perpendicular to the x -axis has diameter $y = x^3$ and area $\frac{\pi}{8} (x^3)^2 = \frac{\pi x^6}{8}$. Therefore, the volume of the solid is given by the definite integral $\int_0^1 \frac{\pi}{8} x^6 dx$, evaluated as follows.

$$\int_0^1 \frac{\pi}{8} x^6 dx = \frac{\pi}{8} \left(\frac{x^7}{7}\right) \Big|_0^1 = \frac{\pi}{56} - 0 = \frac{\pi}{56}$$

8-8-10

42. Let R be the region in the first quadrant, bounded by the graph of $y = 2 \sec x$, the line $x = \frac{\pi}{4}$, and the x - and y -axes. R is the base of a solid whose cross sections perpendicular to the x -axis are semicircles. What is the volume of the solid?

- (A) $\frac{\pi}{2}$
(B) π
(C) 2π
(D) 4π

**Answer A**

Correct. The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. A typical cross section of the solid is a semicircle with diameter from the x -axis to the graph of $y = 2 \sec x$. The length of the diameter is $2 \sec x$; therefore, the radius of the semicircle is $r = \frac{2 \sec x}{2} = \sec x$. The volume of the solid is found using the definite integral of the cross-sectional area as follows.

$$\frac{\pi}{2} \int_0^{\frac{\pi}{4}} \sec^2 x dx = \frac{\pi}{2} \left(\tan x \Big|_0^{\frac{\pi}{4}} \right) = \frac{\pi}{2} \left(\tan\left(\frac{\pi}{4}\right) - \tan(0) \right) = \frac{\pi}{2}$$

43.  The base of a solid is the region in the first quadrant bounded by the graph of $y = \sin x$ and the x -axis for $0 \leq x \leq \pi$. For the solid, each cross section perpendicular to the x -axis is an equilateral triangle. What is the volume of the solid?

- (A) 0.680
(B) 0.866
(C) 1.571
(D) 2.000

**Answer A**

Correct. The area of an equilateral triangle with side y is $\frac{\sqrt{3}}{4}y^2$. At each x , the cross section perpendicular to the x -axis has side $y = \sin x$ and area $\frac{\sqrt{3}}{4}\sin^2 x$. Therefore, the volume of the solid is given by the definite integral

$$\int_0^{\pi} \frac{\sqrt{3}}{4} \sin^2 x dx, \text{ which can be evaluated using the graphing calculator.}$$

8-8-10

44. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = 2x$, the x -axis, and the vertical line $x = 3$ is revolved about the x -axis?
- (A) 9π
- (B) 36π ✓
- (C) 108π
- (D) 324π

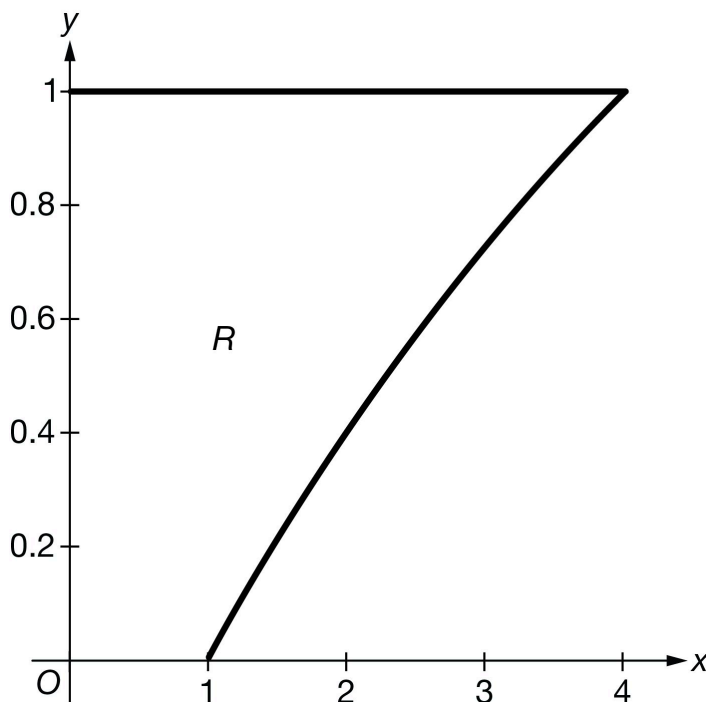
Answer B

Correct. The volume of the solid is given by the definite integral $\pi \int_0^3 (2x)^2 dx$. This is evaluated as follows.

$$\pi \int_0^3 4x^2 dx = \pi \left(\frac{4x^3}{3} \Big|_0^3 \right) = 36\pi - 0 = 36\pi$$

8-8-10

45.



Let R be the region in the first quadrant bounded by the x - and y -axes, the horizontal line $y = 1$, and the graph of $y = \sqrt{x} - 1$, as shown in the figure above. What is the volume of the solid generated when region R is revolved about the y -axis?

- (A) $\frac{7\pi}{3}$
(B) $\frac{31\pi}{5}$
(C) 31π
(D) 155π



Answer B

Correct. Expressed as a function of y , the boundary curve $y = \sqrt{x} - 1$ is $x = (y + 1)^2$ for $0 \leq y \leq 1$. The volume of the solid is given by the definite integral $\pi \int_0^1 x^2 \, dy = \pi \int_0^1 ((y + 1)^2)^2 \, dy$. This is evaluated as follows.

$$\pi \int_0^1 (y + 1)^4 \, dy = \pi \left(\frac{(y+1)^5}{5} \Big|_0^1 \right) = \pi \left(\frac{32}{5} - \frac{1}{5} \right) = \frac{31\pi}{5}$$

8-8-10

46. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = x^3$, the y -axis, and the horizontal line $y = 1$ is revolved about the y -axis?
- (A) $\frac{\pi}{7}$
- (B) $\frac{\pi}{4}$
- (C) $\frac{3\pi}{5}$ ✓
- (D) $\frac{3\pi}{4}$

Answer C

Correct. The volume of the solid is given by the definite integral $\pi \int_0^1 \left(y^{\frac{1}{3}}\right)^2 dy$ and is evaluated as follows.

$$\pi \int_0^1 y^{\frac{2}{3}} dy = \pi \left(\frac{3}{5} y^{\frac{5}{3}} \Big|_0^1 \right) = \frac{3\pi}{5}$$

47. Let R be the triangular region in the first quadrant with vertices at points $(0, 0)$, $(a, 0)$, and $(0, b)$, where a and b are positive constants. Which of the following gives the volume of the solid generated when region R is revolved about the x -axis?
- (A) $\pi \int_0^b \left(\frac{b}{a}x + b \right)^2 dx$
- (B) $\pi \int_0^a \left(\frac{b}{a}x + b \right)^2 dx$
- (C) $\pi \int_0^b \left(\frac{-b}{a}x + b \right)^2 dx$
- (D) $\pi \int_0^a \left(\frac{-b}{a}x + b \right)^2 dx$ ✓

Answer D

Correct. The triangular region is bounded by the x -axis and the line $y = \frac{-b}{a}x + b$ for $0 \leq x \leq a$. Therefore, the volume is given by the definite integral $\pi \int_0^a y^2 dx = \pi \int_0^a \left(\frac{-b}{a}x + b \right)^2 dx$. Evaluating the integral results in

$$\pi \int_0^a \left(\frac{-b}{a}x + b \right)^2 dx = \pi \left(\left(\frac{-a}{b} \right) \left(\frac{1}{3} \right) \left(\frac{-b}{a}x + b \right)^3 \right) \Big|_0^a = \pi \left(0 - \left(\frac{-ab^3}{b(3)} \right) \right) = \frac{1}{3} \pi ab^2$$

Note that the solid generated is a right circular cone with base radius b and height a .

8-8-10

48. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = x^4$, the x -axis, and the vertical line $x = 1$ is revolved about the x -axis?

- (A) $\frac{1}{5}$
(B) $\frac{1}{9}$
(C) $\frac{\pi}{5}$
(D) $\frac{\pi}{9}$

**Answer D**

Correct. The volume of the solid is given by the definite integral $\pi \int_0^1 (x^4)^2 dx$. This is evaluated as follows.

$$\pi \int_0^1 x^8 dx = \pi \left(\frac{x^9}{9} \Big|_0^1 \right) = \frac{\pi}{9}$$

49.  What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = 4\sqrt{4 - x^2}$ and the x - and y -axes is revolved about the y -axis?

- (A) 24.609
(B) 67.021
(C) 268.083
(D) 496.100

**Answer B**

Correct. Expressed in terms of y , the boundary curve $y = 4\sqrt{4 - x^2}$ is $x^2 = \frac{64 - y^2}{16}$ for $0 \leq y \leq 8$. The volume of the solid is given by the definite integral $\pi \int_0^8 x^2 dy = \pi \int_0^8 \left(\frac{64 - y^2}{16} \right) dy$, which can be evaluated with the graphing calculator.

8-8-10

50. Let R be the triangular region in the first quadrant with vertices at points $(0, 0)$, $(h, 0)$, and (h, r) , where r and h are positive constants. Which of the following gives the volume of the solid generated when region R is revolved about the x -axis?

(A) $\pi \int_0^r \left(\frac{h}{r}x\right)^2 dx$

(B) $\pi \int_0^h \left(\frac{h}{r}x\right)^2 dx$

(C) $\pi \int_0^r \left(\frac{r}{h}x\right)^2 dx$

(D) $\pi \int_0^h \left(\frac{r}{h}x\right)^2 dx$

**Answer D**

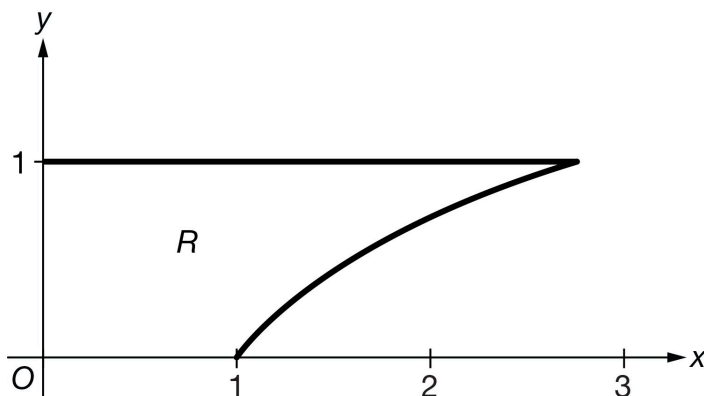
Correct. The triangular region is bounded by the x -axis and the line $y = \frac{r}{h}x$ for $0 \leq x \leq h$. Therefore the volume is given by the definite integral $\pi \int_0^h y^2 dx = \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx$, which is evaluated as follows.

$$\pi \int_0^h \left(\frac{r}{h}x\right)^2 dx = \pi \int_0^h \frac{r^2}{h^2} x^2 dx = \pi \frac{r^2}{h^2} \frac{x^3}{3} \Big|_0^h = \pi \frac{r^2}{h^2} \frac{h^3}{3} = \frac{1}{3} \pi r^2 h$$

Note that the solid generated is a right circular cone with base radius r and height h .

8-8-10

51.



Let R be the region in the first quadrant bounded by the x - and y -axes, the horizontal line $y = 1$, and the graph of $y = \ln x$, as shown in the figure above. What is the volume of the solid generated when region R is revolved about the y -axis?

(A) $\pi(e - 1)$

(B) $\frac{\pi(e^2 - 1)}{2}$



(C) $\pi(e^2 - 1)$

(D) $2\pi(e^2 - 1)$

Answer B

Correct. Expressed as a function of y , the boundary curve $y = \ln x$ is $x = e^y$ for $0 \leq y \leq 1$. The volume of the solid is given by the definite integral $\pi \int_0^1 x^2 \, dy = \pi \int_0^1 (e^y)^2 \, dy$. This is evaluated as follows.

$$\pi \int_0^1 (e^y)^2 \, dy = \pi \int_0^1 e^{2y} \, dy = \pi \left(\frac{e^{2y}}{2} \Big|_0^1 \right) = \pi \left(\frac{e^2}{2} - \frac{1}{2} \right)$$

8-8-10

52.  What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = \sqrt{100 - 4x^2}$ and the x - and y -axes is revolved about the y -axis?
- (A) 359.974
- (B) 523.599
- (C) 1047.198
- (D) 4844.731

Answer B

Correct. Expressed in terms of y , the boundary curve $y = \sqrt{100 - 4x^2}$ is $x^2 = \frac{100-y^2}{4}$ for $0 \leq y \leq 10$. The volume of the solid is given by the definite integral $\pi \int_0^{10} x^2 \, dy = \pi \int_0^{10} \frac{100-y^2}{4} \, dy$, which can be evaluated on the graphing calculator.

53. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = x^3$, the x -axis, and the vertical line $x = 2$ is revolved about the x -axis?
- (A) 4
- (B) $\frac{128}{7}$
- (C) 4π
- (D) $\frac{128\pi}{7}$

Answer D

Correct. The volume of the solid is given by the definite integral $\pi \int_0^2 (x^3)^2 \, dx$. This is evaluated as follows.

$$\pi \int_0^2 (x^3)^2 \, dx = \pi \int_0^2 x^6 \, dx = \pi \left. \frac{x^7}{7} \right|_0^2 = \frac{128\pi}{7}$$

8-8-10

54. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = x^2$, the y -axis, and the horizontal line $y = 1$ is revolved about the y -axis?
- (A) $\frac{\pi}{5}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{\pi}{2}$ ✓
- (D) $\frac{2\pi}{3}$

Answer C

Correct. The volume of the solid is given by the definite integral $\pi \int_0^1 x^2 \, dy = \pi \int_0^1 y \, dy$ and is evaluated as follows.

$$\pi \int_0^1 y \, dy = \pi \left. \frac{y^2}{2} \right|_0^1 = \frac{\pi}{2}$$

55. What is the volume of the solid generated when the region in the first quadrant bounded by the graph of $y = e^x$, the x -axis, and the vertical line $x = \ln 2$ is revolved about the x -axis?
- (A) π
- (B) $\frac{3\pi}{2}$ ✓
- (C) 3π
- (D) 6π

Answer B

Correct. The volume of the solid is given by the definite integral $\pi \int_0^{\ln 2} (e^x)^2 \, dx$. This is evaluated as follows.

$$\pi \int_0^{\ln 2} (e^x)^2 \, dx = \pi \int_0^{\ln 2} e^{2x} \, dx = \pi \left(\frac{e^{2x}}{2} \Big|_0^{\ln 2} \right) = \pi \left(\frac{e^{2 \ln 2}}{2} - \frac{e^0}{2} \right) = \frac{3\pi}{2}$$

8-8-10

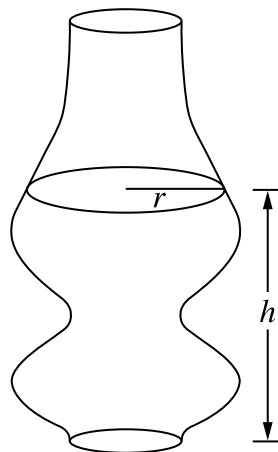
56.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



h (centimeters)	0	6.2	13	20
$r(h)$ (centimeters)	2.5	2.7	4.3	2.6

Juice is sold in 20-centimeter-tall bottles with horizontal cross sections parallel to the base that are circles, as shown in the figure. The radius of the circular cross section at height h above the base of the bottle is given by a differentiable function r , where h and $r(h)$ are measured in centimeters. Selected values of $r(h)$ are given in the table shown.

- (a) Approximate $r'(3.1)$ using the average rate of change of r over the interval $0 \leq h \leq 6.2$. Show the computations that lead to your answer. Indicate units of measure.

8-8-10

(b) Interpret the meaning of $\frac{1}{20} \int_0^{20} r(h) \, dh$ in the context of the problem. Use a trapezoidal sum with the three subintervals $[0, 6.2]$, $[6.2, 13]$, and $[13, 20]$ to approximate the value of $\frac{1}{20} \int_0^{20} r(h) \, dh$.

(c) The radius of the circular cross section at height h above the base of the bottle can also be modeled by the function f , where h and $f(h)$ are measured in centimeters. Let

$f(h) = 3\sqrt{2 - 1.25 \cos\left(\frac{\pi}{100}(h - 20)^2\right)}$ for $0 \leq h \leq 20$. Based on the model, at a height of $h = 12$ centimeters, is the radius increasing or decreasing as h increases? Give a reason for your answer.

(d) Using the model f defined in part (c), find the volume of the bottle.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a difference and a quotient. A response of only $\frac{2.7-2.5}{6.2}$ or $\frac{0.2}{6.2-0}$ earns the first point.

To earn the second point a response must have $\frac{\text{cm}}{\text{cm}}$ (or the equivalent) attached to a numerical value. These units do not reduce to 1 because the numerator, cm, refers to radius of the bottle and the denominator, cm, refers to height of the bottle.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Units

Solution:

$$r'(3.1) \approx \frac{r(6.2) - r(0)}{6.2 - 0} = \frac{2.7 - 2.5}{6.2}$$

0.032 centimeter per centimeter

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-8-10

To earn the first point, a response must provide “average,” “radius,” and “the interval from 0 to 20” (or equivalent such as “throughout the bottle” or “from the bottom to the top”). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the trapezoidal sum must be correct.

If there is any error in the trapezoidal sum, the response does not earn the third point.

A response of $\frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right)$ earns the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that is missing the factor of $\frac{1}{2}$ in the trapezoidal sum does not earn the second and third points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of trapezoidal sum
- ☐ Approximation

Solution:

$\frac{1}{20} \int_0^{20} r(h) \, dh$ is the average value of the radius of the juice bottle, in centimeters, over the interval $0 \leq h \leq 20$ centimeters.

$$\begin{aligned}
 & \frac{1}{20} \int_0^{20} r(h) \, dh \\
 & \approx \frac{1}{20} \left(6.2 \cdot \frac{r(0)+r(6.2)}{2} + 6.8 \cdot \frac{r(6.2)+r(13)}{2} + 7 \cdot \frac{r(13)+r(20)}{2} \right) \\
 & = \frac{1}{20} \left(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2} \right) \\
 & = \frac{1}{20} \cdot 64.07 = 3.2035
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point, a response does not need to provide a value for $f'(12)$; the sign of $f'(12)$ is sufficient.

Any presented value of $f'(12)$ must be correct for the digits presented, up to three decimal places.

If a response presents an incorrect sign or value for $f'(12)$, the response does not earn the second point.

8-8-10

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $f'(12) = -0.0006660598$. If a response presents this value, it earns the first point but is not eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(12)$
- ☐ Answer with reason

Solution:

$$f'(12) = -0.535902$$

At a height of 12 centimeters, the radius is decreasing as h increases because $f'(12) < 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral with an integrand of $(f(h))^2$, regardless of any subsequent errors in presentation of $f(h)$. A response that *begins* with an error in copying $f(h)$ into an integrand of the form $(f(h))^2$ does not earn the first point, but is eligible to earn the second point for a correct answer (1007.112 or 1007.111).

An answer of 320.574π rounds and truncates to 1007.028, and therefore does not earn the second point. Similarly, an answer of 320.573π truncates to 1007.109 and rounds to 1007.110, and therefore does not earn the second point. However, answers of 320.5736π or 320.5737π do round and truncate to the correct answer and earn the second point.

Degree mode: $\pi \int_0^{20} (f(h))^2 dh = 427.508$ (or 427.507) or 136.0799156π . A response that did not earn the second point in part (c) because of degree mode is eligible to earn the answer point in part (d) for this value. Furthermore, in degree mode 136.079π rounds to 427.505 and truncates to 427.504, and therefore does not earn the second point. However, in degree mode 136.080π is accurate to three decimals and is eligible for the second point.



0	1	2
---	---	---

8-8-10

The student response accurately includes **both** of the criteria below.

☐ Integrand

☐ Answer

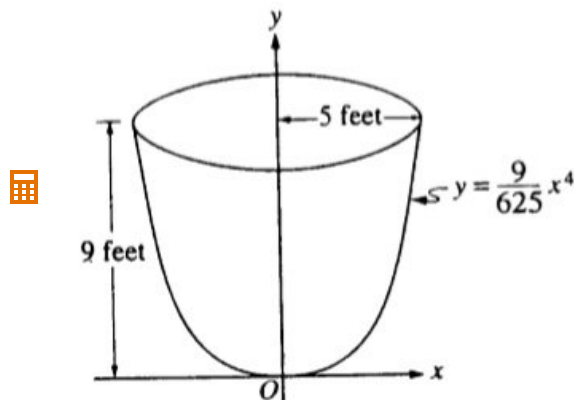
Solution:

$$\text{Volume} = \pi \int_0^{20} (f(h))^2 dh$$

$$= 1007.112 \text{ (or } 1007.111) \text{ cubic centimeters}$$

8-8-10

57.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.

(a) Find the volume of the tank. Indicate units of measure.

(b) To the nearest minute, how long would it take to fill the tank if the tank was empty initially?

(c) Let h be the depth, in feet, of oil in the tank. How fast is the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

Part A

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \pi \int_0^9 () dy, 2\pi \int_0^5 () dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi \text{ ft}^3$$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi \int_0^5 x \left(9 - \frac{9}{625}x^4 \right) dx = 150\pi \text{ ft}^3$$

[465, 475] [148π , 152π]

8-8-10



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \pi \int_0^9 () dy, 2\pi \int_0^5 () dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi = \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi \text{ ft}^3$$

(or 471.238 ft³ or 471.239 ft³)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi = \int_0^5 x \left(9 - \frac{9}{625} x^4 \right) dx = 150\pi \text{ ft}^3$$

[465, 475] [148π, 152π]

Part B

1 point is earned for the correct integer answer rounded up or down

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct integer answer rounded up or down

8-8-10

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure

Part C

1 point is earned for the correct volume as definite integral using h

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct volume as definite integral using h

8-8-10

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$ 1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

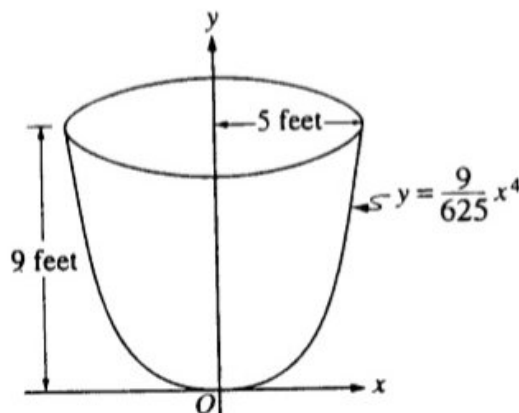
Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)

8-8-10

58.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

- (a) Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.
- (b) Find, to the nearest foot-pound, the amount of work required to empty the tank by pumping all of the oil back to the top of the tank.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} \, dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

8-8-10

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} \, dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

Part B

1 point is earned for the correct limits

$$W = 50 \int_0^6 (9 - y) \left(\frac{25\pi}{3} \sqrt{y} \right) dy$$

8-8-10

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

$$W = 50 \left(\frac{25\pi}{3} \right) \int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}} \right) dy$$

$$W = 50 \left(\frac{25\pi}{3} \right) \left(\frac{2}{3} \cdot 9y^{\frac{3}{2}} - \frac{2}{5}y^{\frac{5}{2}} \right) \Big|_0^6$$

1 point is earned for the correct answer

$W = 69,257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

$$W = 50 \int_0^6 (9 - y) \left(\frac{25\pi}{3} \sqrt{y} \right) dy$$

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

$$W = 50 \left(\frac{25\pi}{3} \right) \int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}} \right) dy$$

$$W = 50 \left(\frac{25\pi}{3} \right) \left(\frac{2}{3} \cdot 9y^{\frac{3}{2}} - \frac{2}{5}y^{\frac{5}{2}} \right) \Big|_0^6$$

1 point is earned for the correct answer

$W = 69,257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)

8-8-10

59.  Let R be the region in the first quadrant bounded by the y -axis, the line $y = 2$, and the curve $y = 2\sqrt[3]{x}$. What is the volume of the solid formed by revolving R about the y -axis?

(A) 0.500

(B) 0.898



(C) 1.571

(D) 23.937

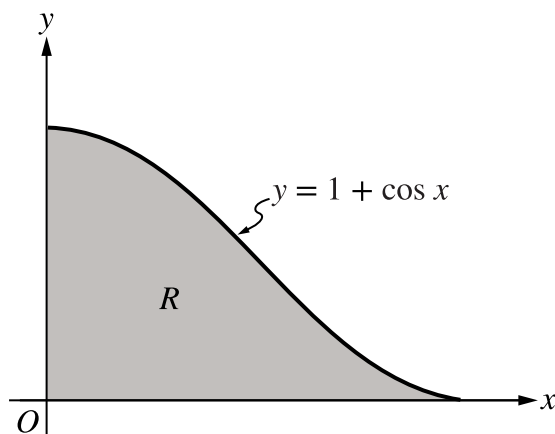
Answer B

Correct. Rotating a typical slice around the y -axis will form a disk with radius r , where the radius must be expressed in terms of y . Since $y = 2\sqrt[3]{x}$, the radius is $r = x = \left(\frac{y}{2}\right)^3$. The cross-sectional area of a typical slice rotated around the y -axis is therefore $A(y) = \pi r^2 = \pi\left(\frac{y}{2}\right)^6$. The volume of the solid is found using the definite integral of the cross-sectional area for y between 0 and 2, as follows.

$$\int_0^2 \pi\left(\frac{y}{2}\right)^6 dy = 0.898$$

8-8-10

60.



 Let R be the shaded region in the first quadrant bounded by the x -axis, the y -axis, and the graph of $y = 1 + \cos x$, as shown in the figure. The shape of a chocolate treat can be modeled by rotating R around the x -axis. What is the volume of the chocolate treat?

- (A) 3.142
(B) 4.712
(C) 9.870
(D) 14.804

**Answer D**

Correct. A typical cross section of the chocolate treat is a disk with radius from the x -axis to the graph of $y = 1 + \cos x$. The area of the disk is $\pi r^2 = \pi(1 + \cos x)^2$. The volume of the treat is found using the definite integral of the cross-sectional area. The lower limit on the definite integral is 0, and the upper limit is π since $1 + \cos x = 0$ when $x = \pi$. The value of the definite integral can be found using the calculator.

$$\int_0^{\pi} \pi(1 + \cos x)^2 dx = 14.804$$

8-8-10

61.  Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the x -axis.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part B

8-8-10

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

8-8-10

$$\begin{aligned}\text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268\end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

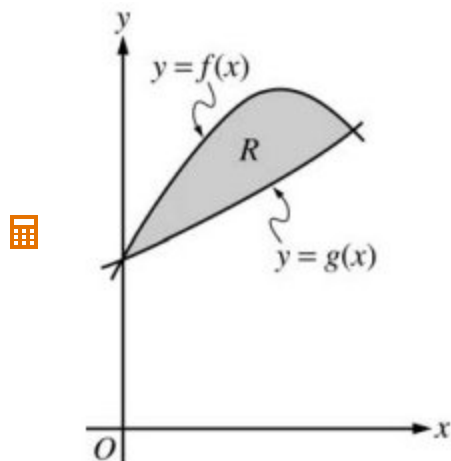
$$\begin{aligned}\text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268\end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

8-8-10

62.



Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles with diameters extending from $y = f(x)$ to $y = g(x)$. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^S (f(x) - g(x)) dx \\
 &= \int_0^S (1 + \sin(2x) - e^{x/2}) dx \\
 &= 0.429
 \end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

8-8-10

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^S (f(x) - g(x))dx \\ &= \int_0^S (1 + \sin(2x) - e^{x/2})dx \\ &= 0.429\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part B

2 point is earned for the integrand

<-1> for each error

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x))dx$

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S ((f(x))^2 - (g(x))^2)dx \\ &= \pi \int_0^S ((1 + \sin(2x))^2 - (e^{x/2})^2)dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point is earned for the integrand

<-1> for each error

8-8-10

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x))dx$

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^S \left((1 + \sin(2x))^2 - (e^{x/2})^2 \right) dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part C

2 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3	4
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The student response earns all of the following points:

2 point is earned for the integrand

1 point is earned for the answer

8-8-10

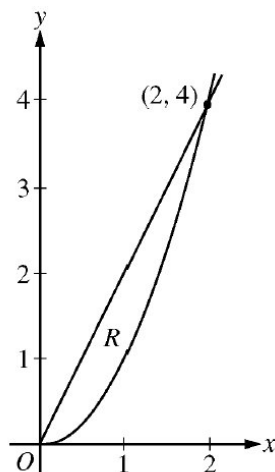
$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

8-8-10

63.



Let R be the region in the first quadrant enclosed by the graphs of $y = 2x$ and $y = x^2$, as shown in the figure above.

- Find the area of R .
- The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \sin(\pi/2x)$. Find the volume of the solid.
- Another solid has the same base R . For this solid, the cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an integral expression for the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^2 (2x - x^2) dx \\
 &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\
 &= \frac{4}{3}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

8-8-10

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3}\end{aligned}$$

Part B

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$

Part C

8-8-10

2 points are earned for the integrand

1 point is earned for the limit

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

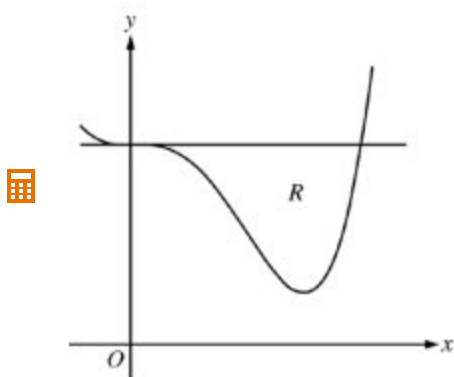
2 points are earned for the integrand

1 point is earned for the limit

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$

8-8-10

64.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

- (a) Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.
- (b) Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in R . Find the volume of the solid.
- (c) The vertical line $x = k$ divides R into two regions with equal areas. Write, but do not solve, an equation involving integral expressions whose solution gives the value k .

Part A

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned} \text{Volume} &= \pi \int_0^{2.3} \left[(4 + 2)^2 - (f(x) + 2)^2 \right] dx \\ &= 98.868 \text{ (or } 98.867) \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

8-8-10

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned}\text{Volume} &= \pi \int_0^{2.3} \left[(4+2)^2 - (f(x)+2)^2 \right] dx \\ &= 98.868 \text{ (or } 98.867\text{)}\end{aligned}$$

Part B

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^{2.3} \frac{1}{2}(4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573\text{)}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^{2.3} \frac{1}{2}(4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573\text{)}\end{aligned}$$

Part C

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

8-8-10



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x))dx = \int_k^{2.3} (4 - f(x))dx$$

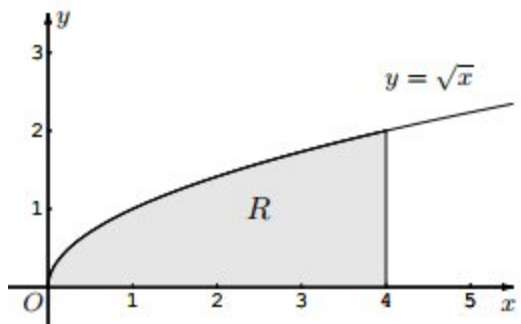
8-8-10

65. Let R be the region bounded by the x -axis, the graph of $y = \sqrt{x}$, and the line $x = 4$.
- Find the area of the region R .
 - Find the value of h such that the vertical line $x = h$ divides the region R into two regions of equal area.
 - Find the volume of the solid generated when R is revolved about the x -axis.
 - The vertical line $x = k$ divides the region R into two regions such that when these two regions are revolved about the x -axis, they generate solids with equal volumes. Find the value of k .

Part A

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation



$$A = \int_0^4 \sqrt{x} dx = \frac{2}{3} x^{3/2} \Big|_0^4 = \frac{16}{3} \text{ or } 5.333$$



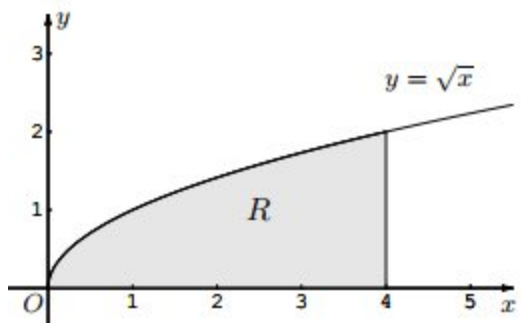
0	1	2
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The student response earns all of the following points:

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation

8-8-10



$$A = \int_0^4 \sqrt{x} \, dx = \frac{2}{3} x^{3/2} \Big|_0^4 = \frac{16}{3} \text{ or } 5.333$$

Part B

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} \, dx = \frac{8}{3} \quad \int_0^h \sqrt{x} \, dx = \int_h^4 \sqrt{x} \, dx$$

or

$$\frac{2}{3} h^{3/2} = \frac{8}{3} \quad \frac{2}{3} h^{3/2} = \frac{16}{3} - \frac{2}{3} h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} \, dx = \frac{8}{3} \quad \int_0^h \sqrt{x} \, dx = \int_h^4 \sqrt{x} \, dx$$

or

$$\frac{2}{3} h^{3/2} = \frac{8}{3} \quad \frac{2}{3} h^{3/2} = \frac{16}{3} - \frac{2}{3} h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$

8-8-10

Part C

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132



0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132

Part D

1 point is earned for equation in k

1 point is earned for the answer

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi \quad \pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

or

$$\pi \frac{k^2}{2} = 4\pi \quad \pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \quad \text{or} \quad 2.828$$

8-8-10



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for equation in k

1 point is earned for the answer

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi \quad \pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

or

$$\pi \frac{k^2}{2} = 4\pi \quad \pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \quad \text{or} \quad 2.828$$

8-8-10

66.  Let R be the region enclosed by the graph of $y = \sqrt{x - 1}$, the vertical line $x = 10$, and the x -axis.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 3$.
- (c) Find the volume of the solid generated when R is revolved about the vertical line $x = 10$.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x - 1} \, dx = 18$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x - 1} \, dx = 18$$

Part B

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x - 1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

8-8-10

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x-1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_0^3 \left(10 - (y^2 + 1) \right)^2 dy$$

$$= 407.150$$

✓

0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

8-8-10

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$= 407.150$$

67.  Let R be the region in the first quadrant bounded by the curve $y = e^{x^2}$, the x - and y -axes, and the vertical line $x = 2$. What is the volume of the solid generated when R is revolved about the x -axis?

- (A) 16.453
(B) 51.687
(C) 403.020
(D) 1266.125



Answer D

Correct. A typical cross section of the solid is a disk with radius from the x -axis to the graph of $y = e^{x^2}$. The area of the disk is $\pi r^2 = \pi(e^{x^2})^2 = \pi e^{2x^2}$. The volume of the solid is found using the definite integral of the cross-sectional area. The lower limit on the definite integral is 0, and the upper limit is 2.

$$\int_0^2 \pi e^{2x^2} dx = 1266.125$$

The graphing calculator is used to evaluate the definite integral.

8-8-10

68.  What is the volume of the solid formed when the region bounded by the graphs of $f(x) = \frac{1}{2}x^2 - x + 1$ and $g(x) = 5 \sin\left(\frac{x}{3}\right)$ is revolved about the x -axis?

- (A) 5.006
(B) 23.815
(C) 26.738
(D) 74.816



Answer D

Correct. The two graphs intersect at $A = 0.40681$ and $B = 3.94532$. In addition, the graphs on the calculator show that $f(x) \leq g(x)$ for $A \leq x \leq B$. Rotating a typical slice of the region about the x -axis will form a washer with outer radius $R(x) = g(x) = 5 \sin\left(\frac{x}{3}\right)$ and inner radius $r(x) = f(x) = \frac{1}{2}x^2 - x + 1$. The cross-sectional area of a typical slice is therefore $A(x) = \pi\left((R(x))^2 - (r(x))^2\right) = \pi\left(25 \sin^2\left(\frac{x}{3}\right) - \left(\frac{1}{2}x^2 - x + 1\right)^2\right)$.

The volume of the solid is found by using the definite integral of the cross-sectional area for x between A and B .

$$V = \pi \int_A^B \left((R(x))^2 - (r(x))^2 \right) dx = \int_{0.40681}^{3.94532} \pi \left(25 \sin^2\left(\frac{x}{3}\right) - \left(\frac{1}{2}x^2 - x + 1\right)^2 \right) dx = 74.816$$

The definite integral is evaluated with the graphing calculator.

69. What is the volume of the solid generated when the region bounded by the graph of $y = 2x$, the horizontal line $y = 2$, and the y -axis is revolved about the horizontal line $y = 2$?

- (A) π
(B) $\frac{4\pi}{3}$
(C) $\frac{8\pi}{3}$
(D) $\frac{16\pi}{3}$



Answer B

Correct. The region is bounded above by the horizontal line $y = 2$ and below by the graph of $y = 2x$ for $0 \leq x \leq 1$. The volume of the solid is given by the definite integral $\pi \int_0^1 (2 - 2x)^2 dx$, which is evaluated as follows.

$$\pi \int_0^1 (2 - 2x)^2 dx = \pi \left(-\frac{1}{6}(2 - 2x)^3 \Big|_0^1 \right) = \pi \left(0 - \left(-\frac{8}{6} \right) \right) = \frac{4\pi}{3}$$

8-8-10

70. Let R be the region in the first quadrant bounded by the graph of $y = \sin x$, the x -axis, and the vertical line $x = 1$. Which of the following gives the volume of the solid generated when region R is revolved about the vertical line $x = 1$?

(A) $\pi \int_0^{\sin 1} (1 - \arcsin y)^2 dy$ ✓

(B) $\pi \int_0^1 (1 - \arcsin y)^2 dy$

(C) $\pi \int_0^{\sin 1} (1 - \arcsin y) dy$

(D) $\pi \int_0^{\sin 1} (1 - (\arcsin y)^2) dy$

Answer A

Correct. Expressed as a function of y , the boundary curve $y = \sin x$ is $x = \arcsin y$ for $0 \leq y \leq \sin 1$. Since the radius of a cross-sectional circle of the solid is $1 - \arcsin y$, the volume is given by the definite integral

$$\pi \int_0^{\sin 1} (1 - \arcsin y)^2 dy.$$

8-8-10

71. Let R be the region bounded by the graph of $y = 2x - 2$, the horizontal line $y = 2$, and the vertical line $x = 1$. Which of the following gives the volume of the solid generated when region R is revolved about the vertical line $x = 1$?

(A) $\pi \int_1^2 \left(\frac{y+2}{2} - 1 \right)^2 dy$

(B) $\pi \int_0^2 \left(\frac{y+2}{2} - 1 \right)^2 dy$ ✓

(C) $\pi \int_1^2 (2 - (2x - 2))^2 dx$

(D) $\pi \int_0^2 \left(\left(\frac{y+2}{2} \right)^2 - 1^2 \right) dy$

Answer B

Correct. Expressed as a function of y , the line $y = 2x - 2$ is $x = \frac{y+2}{2}$. The region is bounded on the right by the graph of $x = \frac{y+2}{2}$ and on the left by the vertical line $x = 1$ for $0 \leq y \leq 2$. Since the radius of a cross-sectional circle of the solid is $\frac{y+2}{2} - 1$, the volume is given by the definite integral $\pi \int_0^2 \left(\frac{y+2}{2} - 1 \right)^2 dy$.

8-8-10

72. Let R be the region enclosed by the graphs of $y = 5x - x^4$ and the horizontal line $y = \frac{3}{2}$. The x -coordinates of the points of intersection of the graphs are x_1 and x_2 , where $x_1 < x_2$. Which of the following gives the volume of the solid generated when region R is revolved about the horizontal line $y = \frac{3}{2}$?

(A) $\int_{x_1}^{x_2} \left(5x - x^4 - \frac{3}{2}\right)^2 dx$

(B) $\int_{x_1}^{x_2} \left((5x - x^4)^2 - \left(\frac{3}{2}\right)^2\right) dx$

(C) $\pi \int_{x_1}^{x_2} \left(5x - x^4 - \frac{3}{2}\right)^2 dx$



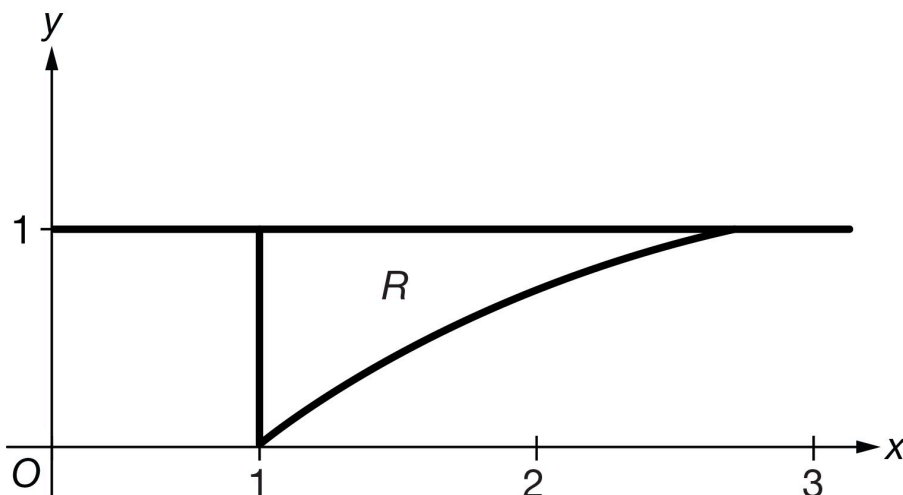
(D) $\pi \int_{x_1}^{x_2} \left((5x - x^4)^2 - \left(\frac{3}{2}\right)^2\right) dx$

Answer C

Correct. For $x_1 \leq x \leq x_2$, the region is bounded above by the graph of $y = 5x - x^4$ and below by the graph of the horizontal line $y = \frac{3}{2}$. The radii of the solid of revolution are given by $(5x - x^4) - \frac{3}{2}$.

8-8-10

73.



Let R be the region bounded by the graph of $y = \ln x$, the horizontal line $y = 1$, and the vertical line $x = 1$, as shown in the figure above. Which of the following gives the volume of the solid generated when the region R is revolved about the vertical line $x = 1$?

(A) $\pi \int_0^1 (e^y - 1) \, dy$

(B) $\pi \int_0^1 (e^{2y} - 1) \, dy$

(C) $\pi \int_0^1 (e^y - 1)^2 \, dy$ ✓

(D) $\pi \int_0^1 (e^y + 1)^2 \, dy$

Answer C

Correct. Expressed as a function of y , the boundary curve $y = \ln x$ is $x = e^y$ for $0 \leq y \leq 1$. Since the radius of a cross-sectional circle of the solid is $e^y - 1$, the volume is given by the definite integral $\pi \int_0^1 (e^y - 1)^2 \, dy$.

8-8-10

74. What is the volume of the solid generated when the region bounded by the graph of $y = x^2$, the vertical line $x = 3$, and the horizontal line $y = 4$ is revolved about the horizontal line $y = 4$?

(A) $\pi \int_2^3 (x^2 - 4) \, dx$

(B) $\pi \int_2^3 (x^4 - 16) \, dx$

(C) $\pi \int_2^3 (x^2 - 4)^2 \, dx$ ✓

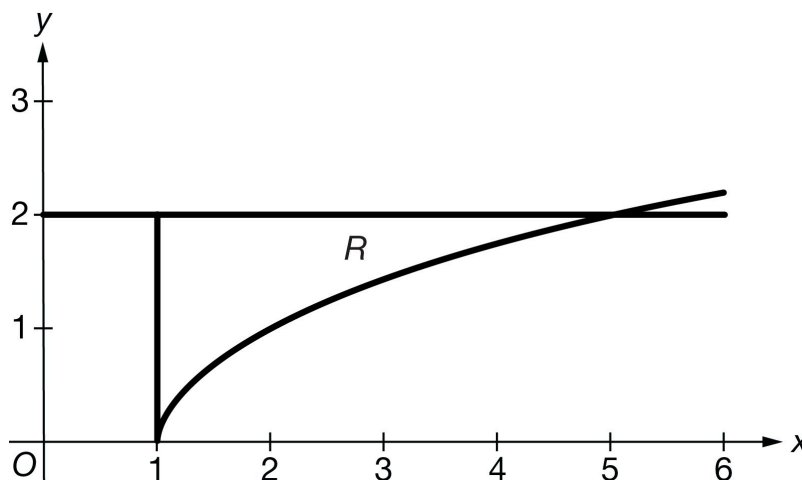
(D) $\pi \int_2^3 (x^2 + 4)^2 \, dx$

Answer C

Correct. The region is bounded above by the graph of $y = x^2$ and below by the horizontal line $y = 4$ for $2 \leq x \leq 3$. Since the radius of a cross-sectional circle of the solid is $x^2 - 4$, the volume is given by the definite integral $\pi \int_2^3 (x^2 - 4)^2 \, dx$.

8-8-10

75.



Let R be the region bounded by the graph of $y = \sqrt{x-1}$, the horizontal line $y = 2$, and the vertical line $x = 1$, as shown in the figure above. Which of the following gives the volume of the solid generated when the region R is revolved about the vertical line $x = 1$?

(A) $\pi \int_0^2 ((y^2 + 1) - 1) \, dy$

(B) $\pi \int_0^2 ((y^2 + 1)^2 - 1^2) \, dy$

(C) $\pi \int_0^2 ((y^2 + 1) - 1)^2 \, dy$ ✓

(D) $\pi \int_0^2 ((y^2 + 1) + 1)^2 \, dy$

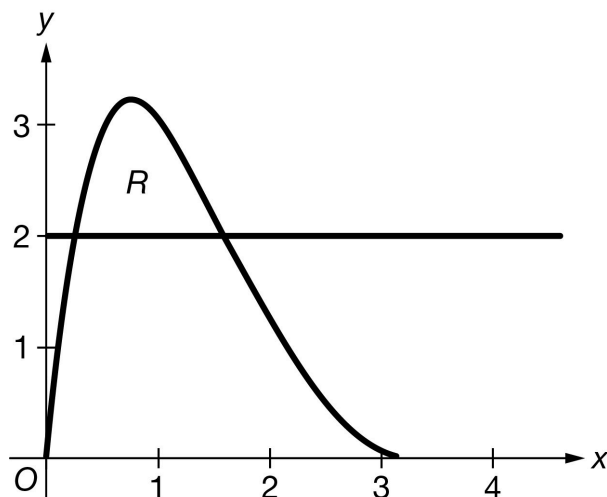
Answer C

Correct. Expressed as a function of y , the boundary curve $y = \sqrt{x-1}$ is $x = y^2 + 1$ for $0 \leq y \leq 2$. Since the radius of a cross-sectional disc (circle) of the solid is $(y^2 + 1) - 1$,

the volume is π times the radius squared, given by the definite integral $\pi \int_0^2 ((y^2 + 1) - 1)^2 \, dy$.

8-8-10

76.



Let R be the region in the first quadrant enclosed by the graphs of $y = 10e^{-x} \sin x$ and the horizontal line $y = 2$, as shown in the figure above. The x -coordinates of the two points of intersection of the graphs are x_1 and x_2 , where $0 < x_1 < x_2$. Which of the following gives the volume of the solid generated when region R is revolved about the horizontal line $y = 2$?

- (A) $\int_{x_1}^{x_2} (10e^{-x} \sin x - 2)^2 dx$
- (B) $\int_{x_1}^{x_2} \left((10e^{-x} \sin x)^2 - 2^2 \right) dx$
- (C) $\pi \int_{x_1}^{x_2} (10e^{-x} \sin x - 2)^2 dx$ ✓
- (D) $\pi \int_{x_1}^{x_2} \left((10e^{-x} \sin x)^2 - 2^2 \right) dx$

Answer C

Correct. For $x_1 \leq x \leq x_2$, the region in the first quadrant is bounded above by the graph of $y = 10e^{-x} \sin x$ and below by the graph of the horizontal line $y = 2$. The radii of the cross-sectional discs (circles) of the solid of revolution are given by $(10e^{-x} \sin x) - 2$.

8-8-10

77. Let R be the region in the first quadrant bounded on the right by the graph of $y = \frac{3}{2}x - 3$, above by the horizontal line $y = 3$, and on the left by the vertical line $x = 2$. Which of the following gives the volume of the solid generated when region R is revolved about the vertical line $x = 2$?

(A) $\pi \int_2^4 \left(\left(\frac{2y+6}{3} \right) - 2 \right)^2 dy$

(B) $\pi \int_0^3 \left(\left(\frac{2y+6}{3} \right) - 2 \right)^2 dy$ ✓

(C) $\pi \int_2^4 \left(3 - \left(\frac{3}{2}x - 3 \right) \right)^2 dx$

(D) $\pi \int_0^3 \left(\left(\frac{2y+6}{3} \right)^2 - 2^2 \right) dy$

Answer B

Correct. Expressed as a function of y , the line $y = \frac{3}{2}x - 3$ is $x = \frac{2}{3}(y + 3) = \frac{2y+6}{3}$. The region is bounded on the right by the graph of $x = \frac{2y+6}{3}$ and on the left by the vertical line $x = 2$ for $0 \leq y \leq 3$. Since the radius of a cross-sectional circle of the solid is $\frac{2y+6}{3} - 2$, the volume is given by the definite integral

$$\pi \int_0^3 \left(\left(\frac{2y+6}{3} \right) - 2 \right)^2 dy.$$

8-8-10

78. What is the volume of the solid generated when the region bounded by the graph of $y = x^3$, the vertical line $x = 4$, and the horizontal line $y = 8$ is revolved about the horizontal line $y = 8$?

(A) $\pi \int_2^4 (x^3 - 8) \, dx$

(B) $\pi \int_2^4 (x^6 - 64) \, dx$

(C) $\pi \int_2^4 (x^3 - 8)^2 \, dx$ ✓

(D) $\pi \int_2^4 (x^3 + 8)^2 \, dx$

Answer C

Correct. The region is bounded above by the graph of $y = x^3$ and below by the horizontal line $y = 8$ for $2 \leq x \leq 4$. Since the radius of a cross-sectional circle of the solid is $x^3 - 8$, the volume is given by the definite integral $\pi \int_2^4 (x^3 - 8)^2 \, dx$.

79. What is the volume of the solid generated when the region bounded by the graph of $y = 4x$, the horizontal line $y = 8$, and the y -axis is revolved about the horizontal line $y = 8$?

(A) 8π

(B) $\frac{128\pi}{3}$ ✓

(C) $\frac{256\pi}{3}$

(D) $\frac{512\pi}{3}$

Answer B

Correct. The region is bounded above by the horizontal line $y = 8$ and below by the graph of $y = 4x$ for $0 \leq x \leq 2$. The volume of the solid is given by the definite integral $\pi \int_0^2 (8 - 4x)^2 \, dx$, which is evaluated as follows.

$$\pi \int_0^2 (8 - 4x)^2 \, dx = \pi \left(\frac{1}{(-4)} \frac{(8 - 4x)^3}{3} \bigg|_0^2 \right) = \pi \left(0 - \left(-\frac{8^3}{12} \right) \right) = \frac{128\pi}{3}$$

8-8-10

80. Let R be the region in the first quadrant bounded by the graph of $y = \tan x$, the x -axis, and the vertical line $x = 1$. Which of the following gives the volume of the solid generated when region R is revolved about the vertical line $x = 1$?

- (A) $\pi \int_0^{\tan 1} (1 - \arctan y)^2 dy$ ✓
- (B) $\pi \int_0^1 (1 - \arctan y)^2 dy$
- (C) $\pi \int_0^{\tan 1} (1 - \arctan y) dy$
- (D) $\pi \int_0^1 (1^2 - (\arctan y)^2) dy$

Answer A

Correct. Expressed as a function of y , the boundary curve $y = \tan x$ is $x = \arctan y$ for $0 \leq y \leq \tan 1$. Since the radius of a cross-sectional circle of the solid is $1 - \arctan y$, the volume is given by the definite integral

$$\pi \int_0^{\tan 1} (1 - \arctan y)^2 dy.$$

81. The base of a solid is the region in the first quadrant bounded by the line $2x + y = 2$, the x -axis, and the y -axis. For the solid, each cross section perpendicular to the x -axis is a semicircle.

Which of the following expressions gives the volume of the solid?

- (A) $\frac{\pi}{2} \int_0^1 (1 - x)^2 dx$ ✓
- (B) $\pi \int_0^1 (1 - x)^2 dx$
- (C) $\frac{\pi}{2} \int_0^1 (2 - 2x)^2 dx$
- (D) $\pi \int_0^1 (2 - 2x)^2 dx$

Answer A

Correct. The cross section perpendicular to the x -axis with x -coordinate x is a semicircle with diameter of length $y = 2 - 2x$. The radius of the semicircle is therefore $r = \frac{y}{2} = 1 - x$. The cross-sectional area of the semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(1 - x)^2$. The volume of the solid is found using the definite integral of the cross-sectional area. The line crosses the x -axis when $2x + 0 = 2$, so at $x = 1$. The volume of the solid is therefore $\int_0^1 \frac{\pi}{2}(1 - x)^2 dx$.